Electricity & Magnetism

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1 Electric Charges and Fields

1.1 Electric Charge and Electric Force

Electric Charge

- **Electric charge** (q or Q) is a fundamental property of matter.
- Electric charge is measured in the SI derived unit *Coulombs* (C).
  - Coulombs are equal to *ampere seconds*, being the amount of charge transferred in a 1 A current in one second ($\mathrm{C} = \mathrm{A} \cdot \mathrm{s}$).
- Electric charge comes from **charge carriers**: protons (p^+) and electrons (e^-)
  - The magnitude of charge of both protons and electrons are equal to the *elementary charge*:

$$\text{Elementary Charge} = e = 1.6 \cdot 10^{-19}\,\mathrm{C}$$
$$|q_{\text{proton}}| = |q_{\text{electron}}| = e.$$

  - Protons p^+ carry a positive charge of $+e$, and electrons e^- carry a negative charge of $-e$.
- Electric charge is a **quantized scalar**.
  - **Quantized:** electric charge can only be integer multiples of the elementary charge e.
    In other words, something can only have a charge of $q = ke$, where $k \in \mathbb{Z}$ and e is the elementary charge.
  - **Scalar:** charge does not have direction, but it can be either positive or negative.

Example:
An object consists of 4 protons and 6 electrons. What is its charge in Coulombs?

$$Q = (n_{\text{protons}} - n_{\text{electrons}})e = (4-6)e = -2e = -2(1.60 \cdot 10^{-19}\,\mathrm{C})$$
$$= -3.20 \cdot 10^{-19}\,\mathrm{C}.$$

Law of Charges
Two charges of opposing signs *attract*, and two charges of the same sign *repel*.

- A **point charge** refers to an object or system which has charge, but has negligible physical size. Point charges are easy to deal with because the direction from the charge to any object solely depends on the position of the object.

Atomic Constants

- Mass of a Proton and Neutron[1]: $m_p \approx m_n \approx 1.67 \cdot 10^{-27}\,\mathrm{kg}$
- Mass of an Electron: $m_e \approx 9.11 \cdot 10^{-31}\,\mathrm{kg}$
- Elementary Charge: $e \approx 1.60 \cdot 10^{-19}\,\mathrm{C}$

Coulomb's Law

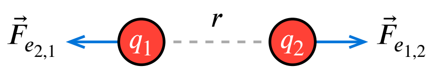
Coulomb's Law

Let q_1 and q_2 be the charges of two point charges a distance r between their centers of charge. Then, the *electrostatic force* $\vec{F}_e$ exerted by q_2 on q_1 is:

$$\text{Electrostatic Force} = \vec{F}_e = k_e \frac{q_1 q_2}{r^2} \hat{r}$$

where:

- k_e is **Coulomb's constant**, defined as $k_e = \frac{1}{4\pi\varepsilon_0} \approx 8.99 \cdot 10^9\,\mathrm{N\,m^2\,C^{-2}}$
  - ε_0 is the **permittivity of free space**, which is a constant which we will define later.
- $\hat{r}$ is the unit vector directed *away* from the location of q_1.
  - in other words, if q_1 and q_2 are the same sign, the force will repel the charges *away* from each other.



- By Newton's Third Law, $\vec{F}_{e_{1,2}} = -\vec{F}_{e_{2,1}}$. Two charges will always exert a pair of electrostatic forces on each other.
- Notice that Coulomb's Law mirrors the Universal Law of Gravitation, $\vec{F}_G = G\frac{m_1 m_2}{r^2}\hat{r}$.
  - $G \approx 6.67 \cdot 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$, the gravitational constant, is *much* smaller than k_e. This means that for large scale objects which are *charged*, gravitational force is negligible compared to the electrostatic force.

    However, it is often the case that large objects *aren't* charged (they are neutral since), which is why we still see the effects of gravitational force.
  - In the case of $\vec{F}_G$, $\hat{r}$ points *towards* the other object rather than *away*. Gravity is always an attractive force. In contrast, electrostatic force can be either attractive or repulsive.

[1] $m_p \neq m_n$, but their difference is negligible.

Example: Relative Electrostatic Force

Two charges of magnitude q placed a distance d from each other exhibit a repulsive force of magnitude F. What is the electrostatic force of two charges of magnitude $3q$ placed a distance $2d$ from each other?

By Coulomb's Law, the initial force is:

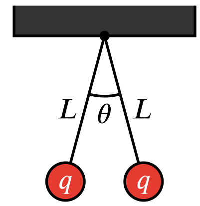
$$F = k_e \frac{q^2}{d^2}.$$

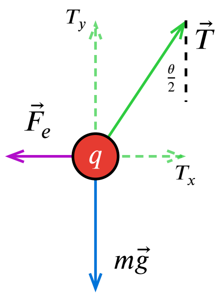
For the new charges, the force is:

$$F_{\text{new}} = k_e \frac{(3q)^2}{(2d)^2} = k_e \frac{9q^2}{4d^2} = \frac{9}{4}F.$$

Example: Charged Bobs on a Pendulum

Two identically charged balls of mass m are in static equilibrium while hanging from the ceiling on strings of length L. The total angle between them is θ ($0° \leq \theta < 180°$). Find the charge q on each ball.





FBD for the left-hanging ball

For vertical forces to balance, set $T_y = mg$:

$$T_y = T\cos\left(\frac{\theta}{2}\right) = mg \implies T = \frac{mg}{\cos\left(\frac{\theta}{2}\right)}.$$

For horizontal forces to balance, set $T_x = F_e$, where F_e is the magnitude of electrostatic repulsion:

$$T_x = T\sin\left(\frac{\theta}{2}\right) = F_e = k_e\frac{q^2}{r^2}.$$

r is the distance between the two balls. Since each ball is a distance $L\sin\left(\frac{\theta}{2}\right)$ from the center, $r = 2L\sin\left(\frac{\theta}{2}\right)$. Thus:

$$T\sin\left(\frac{\theta}{2}\right) = k_e\frac{q^2}{\left(2L\sin\left(\frac{\theta}{2}\right)\right)^2} = k_e\frac{q^2}{4L^2\sin^2\left(\frac{\theta}{2}\right)}.$$

Replace the expression for tension T from earlier and substitute $k_e = \frac{1}{4\pi\varepsilon_0}$:

$$\frac{mg}{\cos\left(\frac{\theta}{2}\right)}\sin\left(\frac{\theta}{2}\right) = k_e\frac{q^2}{4L^2\sin^2\left(\frac{\theta}{2}\right)}$$

$$mg\tan\left(\frac{\theta}{2}\right) = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{4L^2\sin^2\left(\frac{\theta}{2}\right)}.$$

Solving for q:

$$q^2 = 16L^2\sin^2\left(\frac{\theta}{2}\right)\pi\varepsilon_0 mg\tan\left(\frac{\theta}{2}\right)$$

$$q = 4L\sin\left(\frac{\theta}{2}\right)\sqrt{\pi\varepsilon_0 mg\tan\left(\frac{\theta}{2}\right)}.$$

Charge Distributions

Law of Superposition for Coulomb's Law (Discrete)

Let $q_1, q_2, \ldots, q_n$ be a discrete set of charges in space. A particle of charge Q will have its electrostatic force be the vector sum of all individual electrostatic forces:

$$\vec{F}_e = \sum_i \vec{F}_{e_{i,Q}} = \sum_i k_e \frac{q_i Q}{r^2} \hat{r}.$$

In other words, the net electrostatic force is the sum of all individual electrostatic forces.

We can generalize this law to continuous charge distributions. For every small change in charge along a charged object $\mathrm{d}q$, the change in electrostatic force $\mathrm{d}\vec{F}_e$ is:

$$\mathrm{d}\vec{F}_e = k_e \frac{Q\,\mathrm{d}q}{r^2} \hat{r}.$$

Thus, by the law of superposition:

Law of Superposition for Coulomb's Law (Continuous)

Let Q be a point charge and let an object have a continuous charge distribution $\mathscr{D}$. Then, the net electrostatic force $\vec{F}_e$ must be:

$$\vec{F}_e = k_e \int_{\mathscr{D}} \frac{Q}{r^2} \hat{r}\,\mathrm{d}q.$$

1.2 Conservation of Electric Charge

Conservation of Charge

The Law of Conservation of Charge

For isolated systems, net charge is *conserved*:

$$\Sigma Q_i = \Sigma Q_f.$$

This law holds because in an isolated system, charge carriers (i.e. electrons) cannot enter or leave the system.

- Charge cannot be created or destroyed, only *transferred* between objects or particles.
- The *charge distribution* of an isolated system may change, but the *net charge* of the system remains constant.

Insulators and Conductors

- It is difficult to redistribute charges in an **insulator**. Electrons are bound tightly to the atomic nuclei, and cannot easily move. (Electrons are *localized*).
  - We say that the electrons *resist* movement.
  - The degree to which the electrons of a material resist movement is called *resistivity*.
- It is easy to redistribute charges in a **conductor**. Electrons can freely move (they are *delocalized*).
  - The degree to which the electrons of a material are able to move is called *conductivity*. It is the inverse of resistivity (conductivity $= \frac{1}{\text{resistivity}}$).
- **Semiconductors** have properties resembling both conductors and insulators. Electrons are somewhat bound to the atomic nuclei, and conductivity *varies* based on the composition of materials.

Methods of Charging

- An object can have a change in charge when they are introduced into a new system, allowing the charge of the system to distribute into the object.
- When a previously neutral object obtains charge, we say that the object has been *charged*.

Charging by Friction

- When two *insulating* objects are rubbed together, electrons can be transferred from one object to another. This process is called *charging by friction*.
- The object that *gains* electrons becomes *negatively charged*, and the object that *loses* electrons becomes *positively charged*.

Charging by Conduction (Contact)

- When an object touches a *conductor*, charge can be directly transferred through physical contact. The total charge of the object-conductor system is conserved.
- The distribution of charges between the individual objects depend *only* on their **initial charges** and their **sizes (charge capacities)**.
- Two *identical* objects of this manner will evenly distribute charge.
  ex: sphere A ($+4\,\mu\text{C}$) and sphere B ($0\,\mu\text{C}$) make contact with each other.
  If spheres A and B are identical in size, then the final charge on each is $+2\,\mu\text{C}$.
  - To generalize, the final charge of each object after the *conduction* of n identical objects of initial charges $q_1, q_2, \ldots, q_n$ is simply the average charge:

$$q_{\text{individual object after conduction}} = \frac{q_1 + q_2 + \ldots + q_n}{n}.$$

Charging by Induction (Contactless)

- A nearby conductor may *induce* a redistribution of charge in an object. We call this process **induction** (*or* ***polarization***).
  - It is very difficult to induce charge in an insulator, since electrons are not free to move.
- For example, bringing a negatively charged object to the left of a neutral conductor will induce a positive charge on the left side of the conductor and a negative charge on the right side of the conductor.
  - The net charge of the conductor remains *zero* since no charge was added or removed.
  - Taking away the charged object will cause the conductor to return to its original neutral state.

Example: Permanent Induction

Two neutral conducting spheres A and B are placed next to and are touching each other. A negatively charged rod is brought close to sphere A without touching it. After charge distribution reaches equilibrium, sphere B is separated from sphere A. Finally, the rod is removed. Describe the final charges on spheres A and B.

When the negatively charged rod is brought close to sphere A, it induces a positive charge on the left side of sphere A and a negative charge on the right side of sphere A.

Since spheres A and B are touching, the negative charge on the right side of sphere A repels electrons into sphere B, causing sphere B to become negatively charged.

After separation, sphere A retains a net positive charge since it lost electrons to sphere B.

Finally, when the rod is removed, sphere A remains positively charged and sphere B remains negatively charged.

Grounding

- A **ground** $\frac{\underline{\ \ }}{=}$ is a relatively large conductor which can accept or supply an essentially infinite amount of charge.
  - Examples of grounds include the Earth, your body, or a large metal rod connected to the Earth.
- If electrons are able to freely move between a system and a ground, that system is **grounded**. A grounded system is allowed to reach equilibrium by transferring charge between the system and the ground.
  - If a system gains excess charge, that charge is transferred to the ground, allowing the system to return to a neutral state.

1.3 Electric Fields

Electric Field

Let $\vec{E}$ be the net **electric field** at any point P. Then, if a particle with charge q is placed at point P, then the net electrostatic force on the particle is:

$$\vec{F}_e = q\vec{E}.$$

By Coulomb's Law, the electric field at a point P induced by a point charge Q is:

$$\vec{E}_P = \frac{\vec{F}_e}{q} = k_e \frac{Q}{r^2}\hat{r}$$

where $\vec{r}$ is the distance vector from the position of Q to P.

- The **electric field** is the *electrostatic force per unit charge*.
  - Take a test charge q and place it a distance r from another charge Q. The charge Q induces an electrostatic force $\vec{F}_e$ on q and the electric field from Q is $\vec{F}_e/q$.
- The electric field is a vector function of position ($\vec{E}(x, y, z)$), making it a vector field (the electric field at every point is a vector).
- Electric field has units of *newtons per coulomb*, N / C
  - $\mathrm{N\,C^{-1}} = \mathrm{kg\,m\,s^{-3}\,A^{-1}}$ in SI base units.

Electric Field Lines

- Electric field lines generalize the electric field to general regions in space.
- Electric field lines will always move away from positive charges and into negative charges.
- Electric field lines will never cross, and by convention, lines drawn closer together indicate a greater magnitude of electric field.

Electric Fields of Charge Distributions

We can apply the law of superposition for electrostatic force to electric fields.

For discrete charge distributions (a finite set of point charges):

$$\vec{E} = \frac{\vec{F}_e}{Q} = \frac{\sum_i \vec{F}_{e_{i,q}}}{\sum_i q_i} = \sum_i \vec{E}_i.$$

In other words, the net electric field at a given position is the sum of the electric fields induced by all individual point charges.

Often times, we will deal with objects with non-negligible sizes where we must consider the **distribution of charge** throughout the object.

If we split an object into many small, negligibly-sized portions, we can consider each small portion its own point charge that contributes a small amount towards the total electric field.

Let $\mathrm{d}q$ be the additional charge of one of these negligibly-sized point charges. Then the additional electric field $\mathrm{d}\vec{E}$ induced by this point charge is:

$$\mathrm{d}\vec{E} = k_e \frac{\mathrm{d}q}{r^2}\hat{r}.$$

The law of superposition allows us integrate both sides to obtain the net electric field:

$$\vec{E} = k_e \int \frac{1}{r^2}\hat{r}\,\mathrm{d}q.$$

Note

The fact that $\hat{r}$ appears inside our integral can make calculations difficult, since the direction of each $\hat{r}$ can vary as we integrate through different positions along our object.

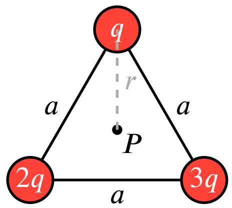
Luckily, at this level, we will only consider scenarios where $\hat{r}$ can be considered constant or where there is symmetry.

Often times, the charge differential $\mathrm{d}q$ is not directly given to us, but rather indirectly using a **charge density**. There are three charge densities:

- **Linear charge density** (λ), charge per unit length.
  - By definition, $\lambda = \frac{\mathrm{d}q}{\mathrm{d}\ell} \implies \mathrm{d}q = \lambda\,\mathrm{d}\ell$ where $\mathrm{d}\ell$ is a small change in length.
  - SI units are *coulombs per meter*, C/m
- **Surface charge density** (σ), charge per unit surface area.
  - By definition, $\sigma = \frac{\mathrm{d}q}{\mathrm{d}A} \implies \mathrm{d}q = \sigma\,\mathrm{d}A$ where $\mathrm{d}A$ is a small change in surface area.
  - SI units are *coulombs per square meter*, $\mathrm{C/m^2}$
- **Volumetric charge density** (ρ), charge per unit volume.
  - By definition, $\rho = \frac{\mathrm{d}q}{\mathrm{d}V} \implies \mathrm{d}q = \rho\,\mathrm{d}V$ where $\mathrm{d}V$ is a small change in volume.
  - SI units are *coulombs per cubic meter*, $\mathrm{C/m^3}$

If charge density is constant, the object is **uniformly charged**.

Example: Finding the Electric Field from Discrete Charges



Three point charges q, $2q$, and $3q$ are arranged in an equilateral triangle of side length a as shown. Find the electric field at the center of the triangle.

The distance r from a vertex to point P is $r = \frac{a}{\sqrt{3}}$. By Coulomb's law, the electric field magnitude due to a charge Q a distance r away is:

$$E = \frac{k_e Q}{r^2} = \frac{k_e Q}{\left(\frac{a}{\sqrt{3}}\right)^2} = \frac{3k_e Q}{a^2}.$$

where k_e is Coulomb's constant.

Thus, the magnitudes of electric field caused by each charge is:

$$E_1 = \frac{3k_e q}{a^2} \qquad E_2 = \frac{6k_e q}{a^2} \qquad E_3 = \frac{9k_e q}{a^2}.$$

By the law of superposition, $\vec{E}_P = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$. However, we only have magnitudes, so we have to use trigonometry to break each magnitude into basis components.

Recall the electric field points *away* from the charge. For example, at point P, $\vec{E}_1$ would point downwards. This means $\vec{E}_1$ has no x-component and a negative y-component, so we derive $\vec{E}_1 = -E_1\hat{\jmath}$.

$\vec{E}_2$ makes a 30° angle with the positive x-axis, so $\vec{E}_2 = E_2(\hat{\imath}\cos 30° + \hat{\jmath}\sin 30°)$. Similarly, $\vec{E}_3$ makes a 30° angle with the *negative* x-axis, so $\vec{E}_3 = E_3(-\hat{\imath}\cos 30° + \hat{\jmath}\sin 30°)$.

Writing out the components:

$$\begin{aligned}
\vec{E}_P &= \left(E_2\cos 30° - E_3\cos 30°\right)\hat{\imath} + \left(-E_1 + E_2\sin 30° + E_3\sin 30°\right)\hat{\jmath} \\
&= \left(E_2 - E_3\right)\frac{\sqrt{3}}{2}\hat{\imath} + \left(-E_1 + \frac{E_2 + E_3}{2}\right)\hat{\jmath} \\
&= \left(-\frac{3k_e q}{a^2}\right)\frac{\sqrt{3}}{2}\hat{\imath} + \left(-3 + \frac{6+9}{2}\right)\frac{k_e q}{a^2}\hat{\jmath} = -\frac{3\sqrt{3}k_e q}{2a^2}\hat{\imath} + \frac{9k_e q}{2a^2}\hat{\jmath}.
\end{aligned}$$

Example: Electric Field from a Continuous Charge Distribution

Point P is a distance b directly to the left of a uniformly charged rod of length L with linear mass density λ. Derive an expression for the electric field at P.

First, since the rod is not a point charge, we are dealing with a *continuous charge distribution*:

$$\vec{E}_P = k_e \int_{x=0}^{x=L} \frac{1}{r^2}\hat{r}\,\mathrm{d}q.$$

Let x be how far from the left side of the rod we are (so we integrate from 0 to L).

Realize that the direction of the electric field is always to the left. So, $\hat{r}$ must be constant and equal to $-\hat{\imath}$ (left-pointing unit vector). Also recognize that r is the distance from P to the left side plus x ($r = b + x$):

$$\vec{E}_P = k_e(-\hat{\imath}) \int_{x=0}^{x=L} \frac{1}{(b+x)^2}\,\mathrm{d}q$$

Since we are given a linear mass density, use $\mathrm{d}q = \lambda\,\mathrm{d}x$:

$$\vec{E}_P = -k_e\hat{\imath} \int_0^L \frac{\lambda}{(b+x)^2}\,\mathrm{d}x.$$

Since we know that the rod is *uniformly charged*, λ is constant, and we can now solve the integral and simplify:

$$\begin{aligned}
\vec{E}_P &= -k_e\lambda\hat{\imath} \int_0^L \frac{1}{(b+x)^2}\,\mathrm{d}x \\
&= -k_e\lambda\hat{\imath}\left[-\frac{1}{b+x}\right]_0^L \\
&= k_e\lambda\left(\frac{1}{b+L} - \frac{1}{b}\right)\hat{\imath} \\
&= -\frac{k_e\lambda L}{b(b+L)}\hat{\imath}.
\end{aligned}$$

Since total charge $Q = \int_0^L \mathrm{d}q = \int_0^L \lambda\,\mathrm{d}x = \lambda L$, we can simplify in terms of Q:

$$\vec{E}_P = -\frac{k_e Q}{b(b+L)}\hat{\imath}$$

1.4 Electric Flux & Gauss's Law

Electric Flux

Let $\vec{E}$ be the electric field and $\mathrm{d}\vec{A}$ be a vector element of area on a surface S. The **electric flux** Φ_E through the surface is:

$$\Phi_E = \iint_S \vec{E} \cdot \mathrm{d}\vec{A}$$

- $\vec{E}$ is the electric field at each point on the surface.
- $\mathrm{d}\vec{A}$ is the vector with its magnitude being a small change in surface area, pointing perpendicularly outwards from the surface
  ($\mathrm{d}\vec{A} = \hat{n}\,\mathrm{d}A$ where $\hat{n}$ is the unit normal to S)

- **Electric flux** measures how much electric field "passes through" a given surface.
  - If $\vec{E}$ is perpendicular to $\mathrm{d}\vec{A}$, the flux is maximized.
  - If $\vec{E}$ is parallel to $\mathrm{d}\vec{A}$, the flux is zero.
  - The dot product ensures that only the component of the electric field perpendicular to the surface contributes to flux.
  - The surface here does not need to be an actual physical surface. It can be an imaginary surface which we use to analyze flux with, called a **Gaussian surface**.
- The symbol Φ_E or φ_E is used to refer to electric flux. It is a scalar quantity.
  (or just Φ if no need to clarify which type of flux)
- The units for electric flux are $\frac{\mathrm{N \cdot m^2}}{\mathrm{C}}$ which is equivalent to the *volt-meter*, $\mathrm{V \cdot m}$.
  - The latter unit is taken directly, *newton-meter squared per coulomb*, $\mathrm{N\,m^2\,C^{-1}}$.
  - A *volt* (V) is a measure of *electrical potential energy per unit charge* (which will be covered later), so volts have units of $\mathrm{J/C} = \mathrm{N\,m\,C^{-1}}$ (unit of energy divided by unit of charge).
  - Thus, a *volt-meter* is equivalent to a *newton-meter squared per coulomb*.
- If electric field is constant everywhere on the surface AND our surface is flat, then our equation simplifies:

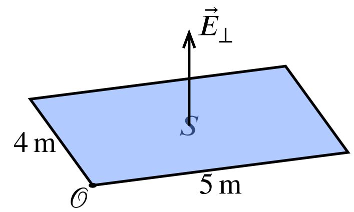
$$\Phi_E = \vec{E} \cdot \iint_S \mathrm{d}\vec{A} = \vec{E} \cdot \vec{A}.$$

- By the definition of the dot product, the electric field for flat surfaces is:
  - $\Phi_E = EA\cos\theta$, if electric field is constant
  - $\Phi_E = \iint_S E\cos\theta\,\mathrm{d}A$, if electric field is not constant

  E is the magnitude of the electric field, A is surface area, and θ is the angle between the outwards-pointing normal vector to the surface and the electric field vector.

Example:

A non-uniform electric field $\vec{E} = \left\langle \alpha xy, \beta z^2, \gamma x + \delta y \right\rangle$ passes through a flat, rectangular surface S as shown in the diagram. What is the electric flux passing through S? Assume x, y, z are specified in meters and $\alpha, \beta, \gamma, \delta$ are constants with appropriate units.



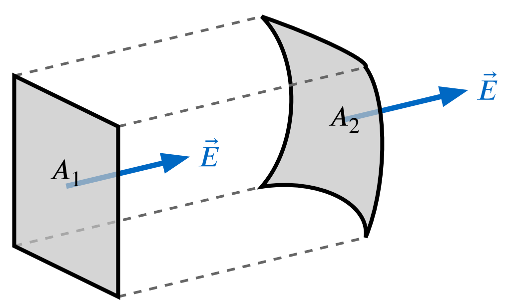
First, since we are only concerned about the component of the electric field that is perpendicular to S, we know that our effective electric field is $E = \gamma x + \delta y$ (i.e. we only consider the z-component of the electric field).

Now, this becomes a simple double integral:

$$\begin{aligned}
\Phi_E &= \iint_S (\gamma x + \delta y)\,\mathrm{d}A \\
&= \int_{y=0}^{4\,\mathrm{m}} \int_{x=0}^{5\,\mathrm{m}} (\gamma x + \delta y)\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^{4\,\mathrm{m}} \left(\frac{\gamma}{2}x^2 + \delta yx\right)\Bigg|_{x=0}^{x=5\,\mathrm{m}} \mathrm{d}y \\
&= \int_0^{4\,\mathrm{m}} \left(25\frac{\gamma}{2} + 5\delta y\right)\mathrm{d}y \\
&= \left(25\frac{\gamma}{2}y + 5\frac{\delta}{2}y^2\right)\Bigg|_0^{4\,\mathrm{m}} = 50\gamma + 40\delta.
\end{aligned}$$

Projection Principle

The *projection principle* states that the electric flux through a curved surface is equal to the electric flux through the projection of that surface onto a plane perpendicular to the electric field.



The electric flux through A_1 is equal to the electric flux through A_2.

If the direction of the electric field is not constant, we can break the curved surface into many small flat surfaces where the electric field is approximately constant, and apply the projection principle to each small surface.

This principle holds because for every small patch of area $\mathrm{d}A$, the contribution to flux is:

$$\mathrm{d}\Phi_E = \vec{E} \cdot \mathrm{d}\vec{A} = E\,\mathrm{d}A\cos\theta = E\,\mathrm{d}A_\perp.$$

Gauss' Law

Assume an isolated point charge of magnitude q is situated at the center of a sphere of radius r. What is the total electric flux over the surface S of the sphere?

Using Coulomb's Law, we see that the magnitude of electric field $\vec{E}$ at any point on the sphere (i.e. a distance r from the charge) is:

$$\left|\vec{E}\right| = k_e \frac{q}{r^2}.$$

Also realize that:

- the magnitude electric field is constant for all points on the surface of the sphere
- the electric field $\vec{E}$ points in the same direction as $\mathrm{d}\vec{A}$ for all points on the surface

Thus, the formula for electric flux simplifies:

$$\Phi_E = \iint_S \vec{E} \cdot \mathrm{d}\vec{A} = \iint_S \left|\vec{E}\right| \mathrm{d}A = E \iint_S \mathrm{d}A = EA$$

Since A would simply be the surface area of the sphere $A = 4\pi r^2$:

$$\Phi_E = EA = \left(k_e \frac{q}{r^2}\right)\left(4\pi r^2\right) = \left(\frac{1}{\cancel{4\pi}\varepsilon_0} \frac{q}{\cancel{r^2}}\right)\cancel{4\pi r^2} = \frac{q}{\varepsilon_0}.$$

It turns out that *all closed surfaces* which enclose a volume ("**Gaussian surfaces**") can be projected as spheres, and thus this formula applies to all such surfaces. Formally:

Gauss' Law (Integral Form)

Let Φ_E be the *electric flux* through a **closed surface** S. Then:

$$\Phi_E = \oiint_S \vec{E} \cdot \mathrm{d}\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0},$$

where:

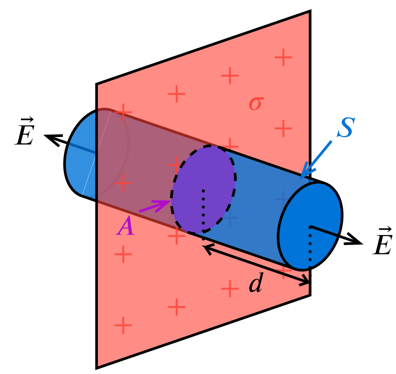
- Q_{enclosed} is the total charge **inside** the volume enclosed by S
- ε_0 is the permittivity of free space.

- The electric flux through any closed surface is proportional to the net charge enclosed within that surface.
- We can choose any closed surface we want (even imaginary ones) to take advantage of this law.
- A common strategy is to choose a Gaussian surface which takes advantage of symmetry to make calculating the electric field easier.
  - Where the electric field is perfectly perpendicular to our surface, then it is parallel to $\hat{n}$ and $\vec{E} \cdot \mathrm{d}\vec{A} = E\,\mathrm{d}A$ (or $-E\,\mathrm{d}A$ if the field points inwards).
  - Where the electric field is perfectly parallel to our surface, then it is perpendicular to $\hat{n}$ and $\vec{E} \cdot \mathrm{d}\vec{A} = 0$.

Example: **Using Gauss' Law to find Electric Field**

An infinite sheet of charge has a uniform surface charge density σ. Use Gauss' Law to derive an expression for the electric field a distance d from the sheet.

The first step is to choose the Gaussian surface.



- The electric field points directly away from the sheet on both sides
- We can choose a cylindrical surface where the electric field pierces through the two circular ends perpendicularly, and is parallel to the curved side of the cylinder. This way, we only have to consider the flux through the two circular ends of the cylinder.

Let the area of each circular end be A, and have the cylinder extend a distance d on either side of the sheet. We can then let the magnitude of electric field at each end be E.

Let's evaluate the left-hand side of Gauss' Law, the flux integral.

The electric field is parallel to the curved side of the cylinder, so they contribute no flux. At each circular end, the electric field is perpendicular to the surface, so the flux through each end is simply EA. Since there are two ends, the total flux is:

$$\Phi_E = EA + EA = 2EA.$$

For the right-hand side, the total charge enclosed by the cylinder is simply the charge on the portion of the sheet inside the cylinder. Since the sheet has a uniform surface charge density σ, the total charge enclosed is:

$$Q_{\text{enclosed}} = \sigma A.$$

Finally, applying Gauss' Law:

$$\begin{aligned} \Phi_E &= \frac{Q_{\text{enclosed}}}{\varepsilon_0} \\ 2EA &= \frac{\sigma A}{\varepsilon_0} \\ E &= \frac{\sigma}{2\varepsilon_0}. \end{aligned}$$

Conductors in Electrostatic Equilibrium

- Remember that in a conductor, electrons are free to move
- If there is an electric field inside the conductor, electrons in the conductor will accelerate.

- For our conductor to be in **electrostatic equilibrium**, charges cannot be accelerating, or in other words there cannot be an electric field.
- Thus, *inside* a conductor at electrostatic equilibrium, $\vec{E} = \vec{0}$.

- By Gauss' Law extra charges in the conductor will be distributed to its surfaces.
  - Choose the Gaussian surface to be any surface completely within the conductor. Since there is no net electric field, there is no electric flux and since $\Phi_E \propto Q_{\text{enc}}$, $Q_{\text{enc}} = 0$.
- External charges can induce polarization on the conducting object, causing one side of the surface to have an opposite charge than the other side.

Introduction to Maxwell's Equations

Gauss' Law is typically the first equation introduced in the set of four equations fundamental to electromagnetism known as *Maxwell's Equations*:

Maxwell's Equations

Name	Integral Form	Differential Form
Gauss' Law	$\oiint_S \vec{E} \cdot \mathrm{d}\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$	$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$
Gauss' Law for Magnetism	$\oiint_S \vec{B} \cdot \mathrm{d}\vec{A} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$
Faraday's Law of Induction	$\oint_C \vec{E} \cdot \mathrm{d}\vec{\ell} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
Ampere-Maxwell Law	$\oint_C \vec{B} \cdot \mathrm{d}\vec{\ell} = \mu_0 I_{\text{enclosed}} + \mu_0 \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t}$	$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

Differential Form of Gauss' Law

Recall the Divergence Theorem:

$$\oiint_S \vec{F} \cdot \mathrm{d}\vec{A} = \iiint_V \vec{\nabla} \cdot \vec{F} \, \mathrm{d}V.$$

So, following Gauss' Law:

$$\begin{aligned} \oiint_S \vec{E} \cdot \mathrm{d}\vec{A} = \iiint_V \vec{\nabla} \cdot \vec{E} \, \mathrm{d}V &= \frac{Q}{\varepsilon_0} \\ \frac{\mathrm{d}}{\mathrm{d}V} \iiint_V \vec{\nabla} \cdot \vec{E} \, \mathrm{d}V &= \frac{\mathrm{d}Q}{\mathrm{d}V} \frac{1}{\varepsilon_0} \\ \vec{\nabla} \cdot \vec{E} &= \frac{\rho}{\varepsilon_0}. \end{aligned}$$

Gauss' Law (Differential Form)

The divergence of the electric field, $\vec{\nabla} \cdot \vec{E}$, at any point is:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

where ρ is volumetric charge density, charge per unit volume ($\mathrm{d}q = \rho \, \mathrm{d}V$).

1.5 Electric Dipoles

- An *electric dipole* consists of two equal and opposite charges of magnitude q separated by a distance d.
- The **electric dipole moment** $\vec{p}$ is a vector that points from the negative charge to the positive charge, with magnitude $p = qd$.
    - Thus, the quantity d is often expressed as $\vec{d}$, with its direction pointing from the negative charge to the positive charge. This way, we can express the dipole moment as:

$$\vec{p} = q\vec{d}.$$

Electric Dipole Moment

Let two charges $+q$ and $-q$ be separated by a distance d. Then the **electric dipole moment** of the system is:

$$\vec{p} = q\vec{d}$$

Such that $\vec{d}$ has magnitude d and points from $-q$ to $+q$.

- An electric dipole is **permanent** if the two charges are fixed in place[2], and **induced** (or *temporary*) if the two charges are free to move and are induced by an external electric field.
- A permanent electric dipole can then be treated as a rigid "rod" which experiences a torque when placed in an external electric field $\vec{E}$. Specifically, the torque $\vec{\tau}$ is given by:

$$\vec{\tau} = \vec{p} \times \vec{E} = q\vec{d} \times \vec{E}.$$

  - Notice that the dipole experiences rotational equilibrium $\tau = 0$ when $\vec{p}$ is parallel, or *aligned with*, the electric field.
  - The potential energy U_p of a dipole in an electric field is given by:

$$U_p = -\vec{p} \cdot \vec{E} = -q\vec{d} \cdot \vec{E}.$$

    It represents the work available to rotate the dipole from its current orientation to a state of zero torque (alignment). Consequently, U_p is lowest when $\vec{p}$ is aligned with $\vec{E}$.

[2]That is, they will always remain a fixed distance d apart

2 Electric Potential and Capacitors

2.1 Electric Potential Energy

Let $\vec{F}_e$ be the electrostatic force on a charge q moving from point A to point B. The **electric potential energy** U_e is defined as the work done by an external agent to move the charge from A to B against the electrostatic force:

$$U_e = W_{\text{external}} = -W_{\text{electrostatic}} = -\int \vec{F}_e \cdot \mathrm{d}\vec{\ell}.$$

If we assume q is only affected by one other point charge Q, we can substitute in Coulomb's Law:

$$U_e = -\int k_e \frac{qQ}{r^2}\hat{r} \cdot \mathrm{d}\vec{\ell}.$$

Realize that $\hat{r}$ and $\mathrm{d}\vec{\ell}$ are always in the same direction (since we are moving along the line between the two charges). Thus, we can let $\mathrm{d}r = \hat{r} \cdot \mathrm{d}\vec{\ell}$:

$$U_e = -k_e qQ \int \frac{1}{r^2}\,\mathrm{d}r.$$

Solving the integral:

$$\begin{aligned} U_e &= -k_e qQ \left[-\frac{1}{r}\right]_{r_A}^{r_B} \\ &= k_e qQ \left(\frac{1}{r_A} - \frac{1}{r_B}\right). \end{aligned}$$

If we let final point be infinitely far away from point A ($r_B \to \infty$), then the electric potential energy at point A is:

$$U_e = k_e \frac{qQ}{r_A}.$$

Thus, we arrive at the general formula for electric potential energy between two point charges:

Electric Potential Energy from a Point Charge

The electric potential energy of a point charge q a distance r from another point charge Q is:

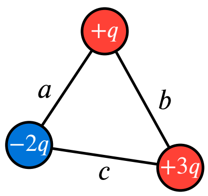
$$U_e = k_e \frac{qQ}{r}.$$

This is the amount of work available to move the charge q from distance r to an infinite distance away from Q.

- Just like all forms of energy, electric potential energy is a *scalar quantity* and is measured in *Joules* (J).

- If both charges have the same sign, then U_e is positive, indicating that work wants to be done to separate the charges (they repel each other).
- If the charges have opposite signs, then U_e is negative, indicating that work is released when the charges come together (they attract each other).
- The reference point used for electric potential energy is that as $r \to \infty, U_e \to 0$.

Example: Electric Potential Energy of a Discrete System



A system of three point charges $+q$, $-2q$, and $+3q$ are arranged in a triangle with side lengths a, b, and c as shown. Calculate the total electric potential energy of the system.

The total potential energy in a system of discrete objects is the sum of potential energies between all pairs of objects.

Between $+q$ and $-2q$: $U_1 = k_e \dfrac{(+q)(-2q)}{a} = -2k_e \dfrac{q^2}{a}$.

Between $+q$ and $+3q$: $U_2 = k_e \dfrac{(+q)(+3q)}{b} = 3k_e \dfrac{q^2}{b}$.

Between $-2q$ and $+3q$: $U_3 = k_e \dfrac{(-2q)(+3q)}{c} = -6k_e \dfrac{q^2}{c}$.

Thus, the total electric potential energy of the system is:

$$U_e = U_1 + U_2 + U_3 = -2k_e \frac{q^2}{a} + 3k_e \frac{q^2}{b} - 6k_e \frac{q^2}{c} = k_e q^2 \left(-\frac{2}{a} + \frac{3}{b} - \frac{6}{c} \right).$$

Example: Electric Potential Energy of a Continuous System

A uniformly charged rod of length L and linear charge density λ lies along the x-axis from $x = 0$ to $x = L$. What is the electric potential energy of a point charge q located a distance d from the left end of the rod on the x-axis?

Since we are dealing with a continuous charge distribution, we use integration to find the total electric potential energy:

$$U_e = k_e q \int_{x=0}^{x=L} \frac{1}{r} \, \mathrm{d}q$$

where r is the distance from the point charge to each small portion of charge on the rod.

Let x be how far from the left side of the rod we are (so we integrate from 0 to L). Then, $r = d + x$ and $\mathrm{d}q = \lambda \, \mathrm{d}x$:

$$U_e = k_e q \int_0^L \frac{1}{d + x} \lambda \, \mathrm{d}x.$$

Since we know that the rod is *uniformly charged*, λ is constant, and we can now solve the integral and simplify:

$$\begin{aligned} U_e &= k_e q \lambda \int_0^L \frac{1}{d + x} \, \mathrm{d}x \\ &= k_e q \lambda \ln(d + x) \Big|_{x=0}^{x=L} \\ &= k_e q \lambda (\ln(d + L) - \ln(d)) \\ &= k_e q \lambda \ln\left(\frac{d + L}{d}\right). \end{aligned}$$

Electric Potential Energy and Electrostatic Force

The electrostatic force is a **conservative force**. The work done by the force over a path only depends on the starting and ending positions, not the path taken.

The change in electric potential energy after following a path C is:

$$\Delta U_e = -\int_C \vec{F}_e \cdot \mathrm{d}\vec{\ell}.$$

By the fundamental theorem of line integrals, we know that if a vector field $\vec{F}$ is the gradient of a scalar field f ($\vec{F} = \vec{\nabla} f$), then:

$$\int_C \vec{F} \cdot \mathrm{d}\vec{\ell} = \int_C \vec{\nabla} f \cdot \mathrm{d}\vec{\ell} = \Delta f.$$

This property only holds for conservative vector fields. Since the electrostatic force is conservative, we can apply this theorem:

$$\int_C \vec{\nabla} U_e \cdot \mathrm{d}\ell = \Delta U_e = \int_C -\vec{F}_e \cdot \mathrm{d}\vec{\ell}.$$

Here, we see that $\vec{\nabla} U_e = -\vec{F}_e$. This is the general definition of electric potential energy:

Electric Potential Energy

The electrostatic force $\vec{F}_e$ on a charge is equal to the negative gradient of its **electric potential energy** U_e:

$$\vec{F}_e = -\vec{\nabla} U_e.$$

In one dimension, this simplifies to:

$$F_{e_x} = -\frac{\mathrm{d}U_e}{\mathrm{d}x}.$$

Things to Recall

- $W = -\Delta U$ ONLY if the work is done by a **conservative force**.
- $\mathrm{ME}_i + W_{\mathrm{nc}} = \mathrm{ME}_f$ in an isolated system
  - The total energy in an isolated system is conserved, just like all other forms of energy:

$$\Delta E = \Delta U + \Delta K = 0 \implies \Delta K = -\Delta U.$$

- $\mathrm{ME}_i + W_{\mathrm{ext}} = \mathrm{ME}_f$ in a non-isolated system

2.2 Electric Potential

Electric Potential from a Point Charge

Let q be a charge in an electric field $\vec{E}$. The electric potential energy U_e of the charge is:

$$U_e = qV$$

where V is the **electric potential** at the position of the charge.

- Electric potential is the electric potential energy per unit charge ($V = U_e/q$).
  - This is similar to how electric field is defined as force per unit charge.
  - Place a charge in a potential field. The potential "energizes" the charge.
- The unit of electric potential is the *volt* (V).
  - A volt is defined as one joule per coulomb (V = J/C).
  - An *electron-volt* (eV) is a unit of *energy*, which is the magnitude of potential energy of an electron in a 1 V potential: $1\text{ eV} = \left(1.6 \cdot 10^{-19}\text{ C}\right)(1\text{ V}) = 1.6 \cdot 10^{-19}\text{ J}$.
- The electric potential V due to a point charge Q at a distance r is:

  $$V = \frac{U_e}{q} = k_e \frac{Q}{r}.$$

  This can be applied to the principle of superposition, where given a collection of point charges $\{q_i\}$,

  $$V = k_e \sum_i \frac{q_i}{r_i}.$$

- Electric potential is a *scalar quantity*.
- Electric potential is often referred to as just "potential".
- An **equipotential line** is a line along which the electric potential is constant.
  - Equipotential lines are always perpendicular to electric field lines (since $\vec{E} = -\vec{\nabla} V$).
  - Electric field lines flow from high potential to low potential
- Since $-\vec{\nabla} U_e = \vec{F}_e$, we can derive a relationship between electric potential and electric field:

  $$\vec{F}_e = -\vec{\nabla}(qV) = -q\vec{\nabla} V \implies \vec{E} = \frac{\vec{F}_e}{q} = -\vec{\nabla} V.$$

Electric Field from Electric Potential

The electric field $\vec{E}$ at any point is equal to the negative gradient of the electric potential V at that point:

$$\vec{E} = -\vec{\nabla}V = \left\langle -\frac{\partial V}{\partial x}, -\frac{\partial V}{\partial y}, -\frac{\partial V}{\partial z} \right\rangle.$$

In one dimension, this simplifies to:

$$E_x = -\frac{\mathrm{d}V}{\mathrm{d}x}.$$

Potential Difference

- The **potential difference** ΔV between two points is the change in electric potential between them:

  $$\Delta V = V_B - V_A.$$

  From this, we can also derive the following relationship:

  $$\Delta U_e = q\Delta V \implies \Delta V = \frac{\Delta U_e}{q}.$$

  The potential difference between two points is the change in electric potential energy per unit charge between those two points.

- The electric field points in the direction of decreasing potential (simply because $\vec{E} = -\vec{\nabla}V$)

Since $\Delta U_e = -\int_C \vec{F}_e \cdot \mathrm{d}\vec{\ell}$ and $\vec{F}_e = q\vec{E}$, we can write:

$$\Delta U_e = -\int_C q\vec{E} \cdot \mathrm{d}\vec{\ell} = -q\int_C \vec{E} \cdot \mathrm{d}\vec{\ell}$$
$$\Delta V = \frac{\Delta U_e}{q} = -\int_C \vec{E} \cdot \mathrm{d}\vec{\ell}.$$

Potential Difference from Electric Field

The potential difference ΔV between two points is equal to the negative line integral of the electric field $\vec{E}$ along any path C between those two points:

$$\Delta V = -\int_C \vec{E} \cdot \mathrm{d}\vec{\ell}.$$

- Since $\vec{E} \cdot \mathrm{d}\vec{\ell} = 0$ when $\vec{E}$ is perpendicular to the path, **equipotential lines are always perpendicular to the electric field**.
  - Similarly, the magnitude potential difference is maximized along the direction of the electric field (positive if against, negative if along).

Electric Potential with Conductors

- Recall that inside a conductor, there is zero net electric field: $\vec{E} = \vec{0}$. Thus, the potential difference between any two points *inside* a conductor is zero: $\Delta V = 0$.
- Since the electric field resides on the surface of a conductor, the electric potential *inside* a conductor is equal to the electric potential on its surface (no ΔV).

2.3 Capacitors and Capacitance

- A **capacitor** $\dashv\vdash$ is a device which "stores" electric charge, creating an electric field that results in a potential difference.
- A capacitor involves two separate conductors separated by an insulating material called a **dielectric**. Charges are allowed to move to the edges of each conductor, charging the ends, creating a potential difference.
- The **capacitance** is the maximum charge each conductor can hold per unit potential difference:

Capacitance

The **capacitance** C of a capacitor is defined as the ratio of the magnitude of charge Q on each conductor to the potential difference ΔV between them:

$$C = \frac{Q}{\Delta V}.$$

- The unit of capacitance is the *farad* (F), where 1 F = 1 C/V.

Parallel Plate Capacitors

Two oppositely charged parallel plates a distance d apart create an approximately uniform electric field $\vec{E}$ between them. Let the surface charge densities of the two plates be $+\sigma$ and $-\sigma$, and let each plate have an area A.

Remember that the electric field from an infinite sheet of charge with surface charge density σ is $\frac{\sigma}{2\varepsilon_0}$. With two oppositely charged sheets, the electric field is double that: $E = \frac{\sigma}{\varepsilon_0}$.

The potential difference ΔV between the two plates is:

$$\Delta V = -\int_C \vec{E} \cdot \mathrm{d}\vec{\ell} = -Ed = -\frac{\sigma d}{\varepsilon_0}.$$

By the definition of capacitance:

$$C = \frac{Q}{\Delta V} = \frac{\sigma A}{\frac{\sigma d}{\varepsilon_0}} = \varepsilon_0 \frac{A}{d}.$$

Cylindrical Capacitors

In a cylindrical capacitor, the two conductors are coaxial cylinders with radii a and b ($a < b$) and length L. Note that the electric field between the cylinders is not uniform.

Remember that both conductors must carry the same charge Q, so we can assume they have the same linear charge density $\lambda = Q/L \to Q = \lambda L$.

Remember the electric field a distance r away from an infinitely long wire is $E(r) = \dfrac{\lambda}{2\pi r\varepsilon_0}$. The potential difference between the inner and outer cylinders is:

$$\Delta V = -\int_C \vec{E} \cdot \mathrm{d}\vec{\ell} = -\int_a^b \frac{\lambda}{2\pi r\varepsilon_0}\,\mathrm{d}r = -\frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right).$$

Finally, by the definition of capacitance:

$$C = \frac{Q}{|\Delta V|} = \frac{\lambda L}{\frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right)} = \frac{2\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)}.$$

Spherical Capacitors

In a spherical capacitor, the two conductors are concentric spheres with radii a and b ($a < b$). The electric field outside a sphere with charge Q is $E(r) = k_e\dfrac{Q}{r^2}$. The potential difference between the inner and outer spheres is:

$$\Delta V = -\int_C \vec{E} \cdot \mathrm{d}\vec{\ell} = -\int_a^b k_e\frac{Q}{r^2}\,\mathrm{d}r = -k_eQ\left(\frac{1}{a} - \frac{1}{b}\right).$$

By the definition of capacitance:

$$C = \frac{Q}{|\Delta V|} = \frac{Q}{k_eQ\left(\frac{1}{a} - \frac{1}{b}\right)} = \frac{4\pi\varepsilon_0}{\frac{1}{a} - \frac{1}{b}}.$$

Energy of Capacitors

Capacitors store electric potential energy. Since charges want to move from one conductor to the other conductor in a capacitor, but can't, they have potential energy.

The **potential energy** stored by a capacitor is the work which can be done to move the charges from one conductor to another through the electric field between the conductors.

To find the potential energy, we can calculate the work done to charge the capacitor. Let Q be the final charge on each conductor, and let q be the charge at some intermediate point in the charging process. The potential difference at that point is $\Delta V = q/C$. The work done to move an infinitesimal amount of charge $\mathrm{d}q$ from one conductor to another is:

$$\mathrm{d}W = \Delta V\,\mathrm{d}q = \frac{q}{C}\,\mathrm{d}q.$$

Integrating from 0 to Q:

$$W = \int_0^Q \frac{q}{C}\,\mathrm{d}q = \frac{Q^2}{2C}.$$

Thus, the potential energy stored by a capacitor is:

$$U_C = \frac{Q^2}{2C}.$$

This can be rewritten as:

$$U_C = \frac{1}{2}Q\Delta V = \frac{1}{2}C(\Delta V)^2.$$

Energy Density

The **energy density** u is the amount of potential energy stored per unit volume:

$$u = \frac{U}{\text{volume}}.$$

For a parallel plate capacitor, the volume between the gap is Ad, so:

$$u = \frac{U_C}{Ad} = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}\frac{\varepsilon_0 A}{d}(Ed)^2 = \frac{1}{2}\varepsilon_0 E^2.$$

Realize that this formula no longer depends on any inherent property of the capacitor, *just* electric field. This means that it holds that *all* types of capacitors, not just parallel plate, have the same formula for energy density. In fact, the formula holds for *all* electric fields.

Example: Energy Density of a Spherical Capacitor

A spherical capacitor consists of two concentric spheres with radii a and b ($a < b$). A potential difference of ΔV is maintained between the two spheres. Derive an expression for the energy density u a distance r from the shared center of the spheres.

The energy density of the electric field is given by $u_e = \frac{1}{2}\varepsilon_0 E^2$.

The electric field E a distance r from the center of the spheres is $E(r) = k_e \dfrac{Q}{r^2}$. We can find Q by using the potential difference between the two spheres:

$$\Delta V = k_e Q\left(\frac{1}{a} - \frac{1}{b}\right) \longrightarrow Q = \frac{\Delta V}{k_e\left(\frac{1}{a} - \frac{1}{b}\right)} = \frac{\Delta V ab}{k_e(a-b)}.$$

Substituting this into the formula for electric field:

$$E(r) = k_e\frac{Q}{r^2} = \cancel{k_e}\frac{\Delta V ab}{\cancel{k_e}(a-b)r^2} = \frac{\Delta V ab}{r^2(a-b)}.$$

Finally, substituting this into the formula for energy density:

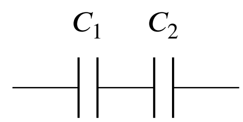
$$u(r) = \frac{1}{2}\varepsilon_0 E^2(r) = \frac{1}{2}\varepsilon_0\left(\frac{\Delta V ab}{r^2(a-b)}\right)^2 = \frac{\varepsilon_0(\Delta V)^2 a^2 b^2}{2r^4(a-b)^2}.$$

Capacitors in Series & Parallel

- When multiple capacitors are placed together in a circuit, their overall effects can be simplified into an **equivalent capacitor**. That is, we can treat groups of capacitors in a circuit as single capacitors.
- The two main arrangements (configurations) of capacitors are **series and parallel**.

Series

In **series**, the capacitors are connected end to end, so **each capacitor has the same charge Q**. However, the potential difference across each capacitor can be different.



Charge is the same on each capacitor due to conservation of charge.

The potential differences would add up to the equivalent potential difference:

$$\Delta V_{\text{eq}} = \Delta V_1 + \Delta V_2 + \ldots + \Delta V_n.$$

By the definition of capacitance, $C = Q/\Delta V \to \Delta V = Q/C$. Substituting this into the sum of potential differences:

$$\Delta V_{\text{eq}} = \frac{Q}{C_{\text{eq}}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \ldots + \frac{Q}{C_n}$$
$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_n}.$$

Parallel

In **parallel**, the capacitors are connected such that **they all share the same potential difference ΔV**, but the charge on each capacitor can be different.

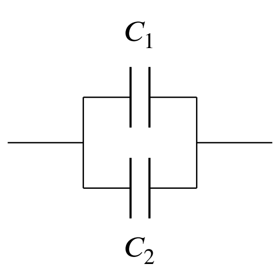
The charges would add up to the equivalent charge:

$$Q_{\text{eq}} = Q_1 + Q_2 + \ldots + Q_n.$$

By the definition of capacitance, $C = Q/\Delta V \to Q = C\Delta V$. Substituting this into the sum of charges:

$$Q_{\text{eq}} = C_{\text{eq}}\Delta V = C_1\Delta V + C_2\Delta V + \ldots + C_n\Delta V$$
$$C_{\text{eq}} = C_1 + C_2 + \ldots + C_n.$$

A parallel connection between C_1 and C_2 looks like the following:



2.4 Dielectrics

- A **dielectric** is an insulating material placed between the conducting plates of a capacitor to *increase* its capacitance.
- When a dielectric is placed in an electric field, it becomes *polarized*, meaning that the positive and negative charges within the dielectric are slightly separated.

  This creates an induced electric field that *opposes* the original electric field, effectively reducing the potential difference between the plates.

  Since potential difference is inversely proportional to capacitance, the capacitance increases.
- Let a capacitor with capacitance C_0 be without a dielectric. When a dielectric is added, the new capacitance is C. The ratio C/C_0 is called the **dielectric constant** of the material, denoted by κ:

$$C = \kappa C_0 \longrightarrow \kappa = \frac{C}{C_0}.$$

  The dielectric constant represents the factor by which capacitance increases when the dielectric is added. For example, with a parallel plate capacitor, the new capacitance with a dielectric is:

$$C = \kappa \varepsilon_0 \frac{A}{d}.$$

  As mentioned, a dielectric really *decreases the potential difference*, so it can be more directly interpreted as the factor by which ΔV is decreased:

$$\Delta V = \frac{1}{\kappa} \Delta V_0.$$

- **Permittivity** is a measure of how easy it is for an electric field to polarize an insulating material.
  - The **permittivity of free space** or **vaccum permittivity** is ε_0.
  - The term $\kappa\varepsilon_0$ is referred to as the **absolute permittivity**, and is denoted with ε.
    - Accordingly, the scaling factor κ is sometimes called **relative permittivity** and denoted ε_r such that $\varepsilon = \varepsilon_r \varepsilon_0$.
  - When talking about conductors, the unit *farad per meter* (F/m) is often used to describe permittivity instead of N m^2/C^2. They are equivalent.

Partial Dielectrics

- A dielectric can partially fill a gap, or multiple types of dielectrics can fill a gap.
  - At the same time, dielectrics can fill gaps discretely or continuously.
- In the discrete case, we can simply treat partial dielectrics as conductors in parallel/series.

Electric Potential Energy in Dielectrics

- If U_0 is the energy stored by a capacitor with capacitance C_0 in a vaccum, then the energy stored by the same capacitor in a material with :

$$U = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}\kappa C_0(\Delta V)^2 = \kappa U_0.$$

- By the same logic, the energy density of an electric field in a dielectric is:

$$u = \frac{1}{2}\varepsilon E^2 = \frac{1}{2}\kappa\varepsilon_0 E^2 = \kappa u_0.$$

Extension: Rigorous Definition of Polarization

- Recall that *electric dipoles* are pairs of equal and opposite charges separated by a small distance, and the *dipole moment* $\vec{p}$ is used to quantify the strength and direction of an electric dipole:

$$\vec{p} = q\vec{d}.$$

- When an external electric field $\vec{E}$ passes through a dielectric material, dipoles within the material align with the field, creating dipole moments throughout the material.

  The **electric polarization** $\vec{P}$, sometimes called the *polarization density*, is then the dipole moment per unit volume:

$$\vec{P} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}V} \quad \rightarrow \quad \vec{p}_{\text{net}} = \iiint_{\text{dielectric}} \vec{P}\,\mathrm{d}V.$$

- The **electric displacement field** $\vec{D}$ is thus the net “effective” electric field:

$$\vec{D} = \varepsilon_0\vec{E} + \vec{P} \quad \rightarrow \quad \vec{E}_{\text{eff}} = \frac{\vec{D}}{\varepsilon_0} = \vec{E} + \frac{1}{\varepsilon_0}\vec{P}.$$

- A dielectric material is **linear** if the polarization $\vec{P}$ is directly proportional to the electric field $\vec{E}$:

$$\vec{P} = \chi_e\varepsilon_0\vec{E}.$$

  The constant of proportionality χ_e is called the **electric susceptibility** of the material, and is a measure of how easily the material becomes polarized in response to an electric field.

  For a linear dielectric, the electric displacement field can be rewritten as:

$$\vec{D} = \varepsilon_0\vec{E} + \chi_e\varepsilon_0\vec{E} = \left(1 + \chi_e\right)\varepsilon_0\vec{E} = \varepsilon\vec{E}.$$

  Here, we see that the absolute permittivity ε is related to the electric susceptibility χ_e by:

$$\varepsilon = \left(1 + \chi_e\right)\varepsilon_0.$$

  Thus, the dielectric constant κ can also be expressed in terms of electric susceptibility:

$$\kappa = \frac{\varepsilon}{\varepsilon_0} = 1 + \chi_e.$$

- At an atomic level, if a single dipole $\vec{p}$ arises in an atom from an electric field $\vec{E}$, then **atomic polarizability** α can be defined as the ratio of the dipole moment to the electric field:

$$\alpha = \frac{p}{E} \quad \rightarrow \quad p = \alpha E.$$

3 Electric Current and Circuits

3.1 Electric Current

Average and Instantaneous Electric Current

Let q be the amount of charge that passes through an arbitrary surface in a time interval Δt. The **average electric current** I_{avg} is:

$$I_{\text{avg}} = \frac{q}{\Delta t}.$$

The **instantaneous electric current** I is the time-derivative of flowing charge:

$$I = \frac{\mathrm{d}q}{\mathrm{d}t}.$$

- Electric current can be thought of as the rate of flow of *positive* electric charge.
  - But electrons are the charge carriers that actually moving, and they are negative. So, electric current is the *negative* rate of flow of electrons.
- Often, current is interpreted as the charge that moves through the cross section of a conductor, since charges can't really "flow" through nonconductors anyways.
- The SI unit of electric current is the *Ampere* (A, "amp"), where 1 A = 1 C/s.
  - In fact, the Ampere is an SI *base unit*, and Coulombs are defined in terms of Amperes (C = A · s).

Current Density

- The **current density** $\vec{J}$ is describes the *electric current per unit area*: $J = I/A$.
- **Current density is a vector** which points in the direction of positive current flow.
- Combining these properties, we can roughly see that $\vec{J} = \frac{\mathrm{d}I}{\mathrm{d}A}\hat{v}$, where $\hat{v}$ is a unit vector in the direction of current flow. Formally:

Electric Current (Current Density Definition)

Let I be the electric current through a surface S. Then:

$$I = \iint_S \vec{J} \cdot \mathrm{d}\vec{A}$$

where:

- $\vec{J}$ is the **current density** vector, which points in the direction of current flow and has magnitude equal to the current per unit area ($J = \frac{\mathrm{d}I}{\mathrm{d}A}$).
- $\mathrm{d}\vec{A}$ is the area vector differential, which points perpendicular to the surface and has magnitude equal to the area of the differential element.

- Current density is measured in *amperes per square meter* ($\mathrm{A/m^2}$).

Drift Velocity

- Each charge carrier (e.g. electron) has a certain **drift velocity** $\vec{v}_d$ which is the average velocity of the charge carriers in the direction of current flow. The current density can also be expressed as:

  $$\vec{J} = nq\vec{v}_d$$

  where n is the number of charge carriers per unit volume, and q is the charge of each carrier.
  It can be interpreted that $\rho = nq$, where ρ is volumetric charge density. However, since charges are moving, we can only give an "average" charge carrier density which we call n.

  - Drift velocities are often very small (on the order of 10^{-4} m/s), but the electric field that causes the drift propagates at nearly the speed of light, so the effect of turning on a current is almost instantaneous.
  - The current density is proportional to drift velocity by a factor of nq.

- By the definition of current density, we can also express current as:

  $$I = \iint_S \vec{J} \cdot \mathrm{d}\vec{A} = \iint_S nq\vec{v}_d \cdot \mathrm{d}\vec{A}.$$

  In a wire, the current density is usually uniform across the cross-sectional area, and the drift velocity is parallel to the $\mathrm{d}\vec{A}$ (i.e. we are taking a perpendicular cross section), so this simplifies to:

  $$I = nqv_d A.$$

Continuity Equation

The **continuity equation** follows directly from conservation of electric charge. Consider an arbitrary closed volume V bounded by a surface S. The total charge enclosed is $Q = \iiint_V \rho \ \mathrm{d}V$.

Any decrease in Q must appear as current flowing outward through S:

$$\oiint_S \vec{J} \cdot \mathrm{d}\vec{A} = -\frac{\mathrm{d}Q}{\mathrm{d}t} = -\frac{\mathrm{d}}{\mathrm{d}t} \iiint_V \rho \ \mathrm{d}V = -\iiint_V \frac{\partial \rho}{\partial t} \ \mathrm{d}V.$$

Applying the Divergence Theorem to the left side:

$$\iiint_V \left(\vec{\nabla} \cdot \vec{J}\right) \mathrm{d}V = -\iiint_V \frac{\partial \rho}{\partial t} \ \mathrm{d}V.$$

So, the integrands must be equal:

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}.$$

Continuity Equation

Let ρ be the volumetric charge density and $\vec{J}$ be the current density. Conservation of charge requires:

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}.$$

An outward divergence of current at a point must be accompanied by a decrease in charge density there, and vice versa.

- In **steady state**, $\partial\rho/\partial t = 0$ everywhere, so $\vec{\nabla} \cdot \vec{J} = 0$: current lines neither originate nor terminate inside a conductor. Current flows continuously around closed loops.
- In integral form: $I_S = -\frac{\mathrm{d}Q_{\text{enc}}}{\mathrm{d}t}$: the net current leaving a closed surface S equals the rate at which the charge enclosed by S is decreasing.

3.2 Electric Circuits

- An **electric circuit** is a closed loop of conductive material through which electric current can flow.
  - This conductive material is called a **wire** and along an <u>isolated</u> piece of wire, electric potential is constant (this makes sense because $\Delta V = 0$ inside a conductor)
- The **circuit elements** are the components of a circuit that affect the flow of current, such as resistors, capacitors, inductors, and voltage sources.
  - Circuit elements have the ability to create potential differences, and thus an electric field within wires to move charges.

Voltage Sources & Electromotive Force

- A **voltage source** is a circuit element that creates a potential difference ΔV between the two **terminals**. The side with the higher potential is usually referred to as the **positive terminal**, and the other end is called the **negative terminal**.
  - An example of a voltage source is a **battery** ┤├, which maintains the potential difference via a chemical reaction.
    - In a circuit diagram, the positive terminal of a voltage source is usually denoted by a longer line, while the negative terminal is denoted by a shorter line.
    - A battery may consist of multiple **cells** connected together, where each cell contributes a certain amount of potential difference. For example, a 9V battery may consist of six 1.5V cells connected in series. **Multicell batteries** are denoted by multiple pairs of lines, e.g. ┤|├ for two cells or ┤||├ for three cells.
  - By the conservation of charge, the same amount of charge pushed through one terminal must eventually return back on the other terminal.

Electromotive Force

The **electromotive force** (emf) $\mathcal{E}$ of a voltage source is the amount of work done by the source to move a unit charge from the negative terminal to the positive terminal:

$$\mathcal{E} = \frac{W}{q}.$$

It is the "ideal" potential difference that the voltage source would create if no current were flowing (i.e. if there was no internal resistance).

- The emf is a measure of the energy provided by the voltage source per unit charge.
- The emf of a voltage source is the voltage across the two terminals when *no current is flowing*.
  - When current flows, some energy is lost due to heat (internal resistance), meaning the actual effective potential difference across the terminals is less than the emf.
- Electromotive force is also measured in Volts (energy per unit charge)!

3.3 Ohm's Law and Resistors

- **Conductivity** σ is a measure of how easily electric current can flow through a material. The higher the conductivity, the easier it is for current to flow.
- The **resistivity** ρ is the reciprocal of conductivity ($\rho = 1/\sigma$) and is a measure of how much a material resists the flow of electric current.

Ohm's Law (Microscopic)

Let $\vec{J}$ be the current density through a material with conductivity σ, and let $\vec{E}$ be the electric field within that material. Then:

$$\vec{J} = \sigma \vec{E} \quad \longleftrightarrow \quad \vec{E} = \rho \vec{J}.$$

The current density within a wire is proportional to the magnitude of the electric field by a factor σ called the ***conductivity****.*

- Ohm's Law tells us that an electric field will drive current through a material, and more conductive materials will allow more current to flow per unit electric field.
  - Conversely, a material with higher resistivity will require a stronger electric field to drive the same amount of current.
- We can find the conductivity or resistivity of a material by comparing the current density and electric field within the material:

$$\sigma = \frac{J}{E} \quad \longleftrightarrow \quad \rho = \frac{E}{J}.$$

- Certain materials have a resistivity that changes with temperature. For example, the resistivity of a metal increases with temperature, while the resistivity of a semiconductor decreases with temperature.
  - Take a material with resistivity ρ_0 at some temperature T_0. If the temperature changes to T and the new resistivity is ρ, then ρ follows the following relation:

$$\rho = \rho_0\big(1 + \alpha\big(T - T_0\big)\big).$$

  We call α the **temperature coefficient of resistivity** of the material.

Resistance

From the equation $E = \rho J$, we can derive the concept of **resistance** R for a wire. Consider a wire with length L and cross-sectional area A. The current density is $J = I/A$, and the electric field is $E = \Delta V/L$. Substituting these into the equation:

$$\frac{\Delta V}{L} = \frac{\rho I}{A} \quad \to \quad \Delta V = \frac{\rho L}{A} I.$$

We can then define $R = \rho L/A$:

Resistance

The **resistance** R of a wire describes how much the wire resists the flow of electric current. It depends on the resistivity ρ of the wire's material, the wire's length L, and cross-sectional area A:

$$R = \frac{\rho L}{A} = \frac{L}{\sigma A}.$$

- That is, the longer wires have more resistance and thicker wires have less resistance.
- Substituting R back into the equation used to derive it, we get $\Delta V = RI$. This is another form of Ohm's Law, measuring the voltage required to drive a unit of current through the wire.

Ohm's Law (Macroscopic)

The current I flowing across a wire with voltage ΔV across it is given by **Ohm's Law**:

$$I = \frac{\Delta V}{R} \quad \longleftrightarrow \quad \Delta V = IR \quad \longleftrightarrow \quad R = \frac{\Delta V}{I}.$$

- The unit of resistance is the *ohm* (Ω), where $1\Omega = 1$ V/A.
  - From this definition, we get the units for resistivity and conductivity: $\Omega\,\text{m}$ and $\frac{1}{\Omega\,\text{m}}$ respectively.
- **Conductance** G is the reciprocal of resistance ($G = 1/R$) and is a measure of how easily electric current can flow through a wire:

$$I = G\Delta V \quad \longleftrightarrow \quad \Delta V = \frac{I}{G} \quad \longleftrightarrow \quad G = \frac{I}{\Delta V}.$$

Internal Resistance & Terminal Voltage

- Batteries in real life have an **internal resistance** r. This means that the battery itself resists the flow of current, and thus the potential difference across the terminals is not equal to the emf $\mathcal{E}$ of the battery. The true potential difference across the terminals, called the **terminal voltage**, is:

$$\Delta V = V_{\text{terminal}} = \mathcal{E} - Ir.$$

  - We can technically represent internal resistance by coupling together an ideal battery and a resistor with resistance r. The "fake" resistor is drawn on the side of the *positive* terminal.
  - An **ideal battery** is one without internal resistance.

> **Note**
> While resistivity and conductivity are inherent properties of *materials*, resistance and conductance further depend on geometric properties of the wire.
>
> For example, while the *resistivity* of copper is always the same, a long thin copper wire will have more *resistance* than a short thick copper wire.

Resistors

- A **resistor** is a circuit element that has a specific resistance R and is used to control the flow of electric current in a circuit.
- Only *some* resistors obey Ohm's Law, meaning that current through the resistor increases linearly with supplied voltage (i.e. $I \propto \Delta V$, and we call the constant of proportionality the resistance R). In other words, the resistance is constant, no matter the voltage supplied.
  - A resistor that obeys Ohm's Law is an **ohmic** resistor. Otherwise, it is a **non-ohmic** resistor.
  - Non-ohmic resistors have a resistance that changes with the voltage across them. For example, a light bulb is a non-ohmic resistor because its filament heats up as current flows through it, which increases its resistance.
  - To check if a resistor is ohmic, check if $\mathrm{d}V/\mathrm{d}R =$ some constant I.

Resistors in Series

Resistors placed in series have their potential differences add up, which means that:

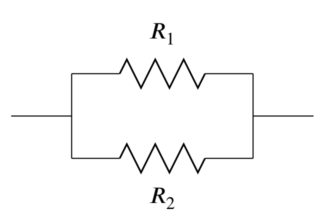
$$\begin{aligned} \Delta V_{\mathrm{eq}} = I R_{\mathrm{eq}} = \Delta V_1 + \Delta V_2 + \ldots + \Delta V_n &= I R_1 + I R_2 + \ldots + I R_n \\ R_{\mathrm{eq}} &= R_1 + R_2 + \ldots + R_n. \end{aligned}$$

In series, resistors share the same current but have different potential differences.

Resistors in Parallel

Resistors placed in parallel have their currents add up but share the same potential difference, which means that:

$$\begin{aligned} I_{\mathrm{eq}} &= I_1 + I_2 + \ldots + I_n \\ \frac{\Delta V}{R_{\mathrm{eq}}} &= \frac{\Delta V}{R_1} + \frac{\Delta V}{R_2} + \ldots + \frac{\Delta V}{R_n} \\ \frac{1}{R_{\mathrm{eq}}} &= \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}. \end{aligned}$$



In parallel, resistors share the same potential difference but have different currents.

Electrical Measuring Instruments

- An **ammeter** —Ⓐ— is a tool used to measure a current at a specific point in a circuit.
  - Ammeters should be connected in series with the circuit element whose current we are measuring
  - Ideal ammeters have zero resistance (such that there is no current drop)
- A **voltmeter** —Ⓥ— is a tool used to measure the potential difference between two points in a circuit.
  - Voltmeters should be connected in parallel with the circuit element across which we are measuring the potential difference.
  - Ideal voltmeters have infinite resistance (such that no current wants to flow through them)
- An **ohmmeter** —Ⓞ— is a tool used to measure the resistance of a circuit element. It works by sending a small current through the element and measuring the potential difference across it, then using Ohm's Law to calculate the resistance.
  - Effectively, an ohmmeter is a combination of an ammeter and a voltmeter.

Microscopic Derivation of Conductivity

In a conductor, free electrons are constantly colliding with the lattice ions of the material, which causes them to lose momentum and change direction. When an electric field is applied, it exerts a force on the electrons, causing them to accelerate between collisions. The average time between collisions is called the **relaxation time** τ.

The average drift speed of the electrons can be derived using Newton's Second Law:

$$ma = m\frac{\mathrm{d}v_d}{\mathrm{d}t} = qE + \frac{mv_d}{\tau}.$$

In steady state, $\mathrm{d}v/\,\mathrm{d}t = 0$, so we can solve for the average drift speed:

$$v_d = \left|-\frac{q\tau}{m}E\right| = \frac{q\tau}{m}E.$$

Substituting this into the equation for current density, we get:

$$J = nqv_d = nq\left(\frac{q\tau}{m}E\right) = \frac{nq^2\tau}{m}E.$$

Thus, the conductivity of the material is:

Conductivity (Microscopic)

The **conductivity** σ of a material with charge carrier density n, charge per carrier q, mass m, and relaxation time τ, the conductivity is given by:

$$\sigma = \frac{nq^2\tau}{m}.$$

3.4 Electric Power

- Recall that mechanical **power** P is the rate at which work is done: $P = \mathrm{d}W/\mathrm{d}t$.
- **Electrical power** P_e is the rate at which electrical energy is transferred by an electric circuit:

$$P_e = \frac{\mathrm{d}W_e}{\mathrm{d}t} = -\frac{\mathrm{d}U_e}{\mathrm{d}t}.$$

  - The electrical power delivered to a circuit element is the rate at which the element "absorbs" electrical energy, causing the element's electric potential energy to decrease.
- We know that when charges move through a potential difference ΔV, they lose potential energy according to $\Delta U_e = q\Delta V$. In other words, the electric field does $-q\Delta V$ amount of work on the charges.

  Thus, the power delivered to a circuit element is:

$$P_e = \frac{\mathrm{d}W_e}{\mathrm{d}t} = -\frac{\mathrm{d}U_e}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(q\Delta V) = -I\Delta V.$$

- If a circuit element is *supplying* power, then the power is positive, so:

Electrical Power

The **electrical power** P_e supplied by a circuit element is given by:

$$P_e = I\Delta V.$$

Note that ΔV is the potential difference across the circuit element, **not** the emf!

- Using Ohm's Law, we can also express power in terms of current and resistance:

$$\begin{aligned} P_e = I\Delta V = I(IR) &= I^2 R. \\ = \left(\frac{\Delta V}{R}\right)\Delta V &= \frac{(\Delta V)^2}{R}. \end{aligned}$$

3.5 Kirchhoff's Rules

Kirchhoff's Loop Rule

Kirchhoff's Loop Rule is a statement about conservation of energy. Because the electric force is conservative, if a charge starts and ends at the same place in a circuit, it must have the same amount of mechanical energy: $\Delta U_e = 0$.

This means that the sum of the potential differences across all circuit elements in a *closed* loop is zero:

$$\sum_i \Delta V_i = 0.$$

This is simply an application of the FTC for line integrals:

$$\oint_C \mathrm{d}V = -\oint_C \vec{E} \cdot \mathrm{d}\vec{r} = -\oint_C \vec{\nabla} V \cdot \mathrm{d}\vec{r} = V(\vec{r}) - V(\vec{r}) = 0.$$

In this case, it is important to keep track of the signs of ΔV_i. Sometimes, voltage *gains* (from emfs like batteries) are separated from voltage *drops* (from resistors and capacitors):

$$\sum_i \Delta V_i = \sum_i \Delta V_{\text{gain}} - \sum_i \Delta V_{\text{drop}} = 0 \quad \rightarrow \quad \sum_i \Delta V_{\text{gain}} = \sum_i \Delta V_{\text{drop}}.$$

The sum of all voltage gains is equal to the sum of all voltage drops.

Again, circuit components can cause either voltage gains or drops:

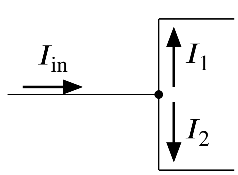
- Passing from the negative terminal to the positive terminal of a voltage source causes a voltage gain, while passing from the positive terminal to the negative terminal causes a voltage drop.
- Passing through a resistor or capacitor in the direction of current flow causes a voltage drop
  - If we pass through a resistor or capacitor *against* the direction of current flow, then it causes a voltage gain.

Kirchhoff's Junction Rule

Kirchhoff's Junction Rule is a statement about conservation of charge. The total current flowing into a junction must equal the total current flowing out of the junction:

$$\sum_i I_{\text{in}} = \sum_i I_{\text{out}}.$$

A **junction** is a point in a circuit where charges have to possibility of taking more than one path.

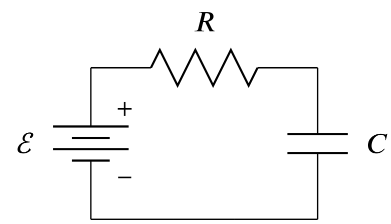


In this example, the dot represents a junction where the current I_{in} splits into two currents I_1 and I_2. By Kirchhoff's Junction Rule, we have $I_{\text{in}} = I_1 + I_2$.

3.6 Resistor-Capacitor (RC) Series Circuits

Charging an RC Series Circuit

An **RC series circuit** is a circuit which is equivalent to one with a resistor and a capacitor in series.



Let's assume that before we close the circuit, the capacitor is uncharged. We can assign $q(t)$ to represent the charge on the capacitor, so $q(0) = 0$.

When we close the circuit, the battery will start pushing charge through the circuit, and the capacitor will start charging up. We call this **charging the capacitor**.

As the capacitor charges, it creates a voltage drop across itself, which reduces the current flowing through the circuit. **Eventually, the capacitor will become fully charged and the current will stop flowing.** That is, the capacitor will act as an open wire once it has reached capacity!

By Kirchhoff's Loop Rule, we can write an equation for the circuit:

$$\mathcal{E} - \Delta V_R - \Delta V_C = 0 \implies \mathcal{E} = \Delta V_R + \Delta V_C.$$

We closed the switch, so there should be some current $i = \mathrm{d}q/\,\mathrm{d}t$. The voltage drop across the resistor is $V_R = IR$, and the voltage drop across the capacitor is $V_C = q/C$. Substituting these into the equation, we get a first-order differential equation which allows us to solve for $q(t)$:

$$\mathcal{E} = iR + \frac{q}{C} \quad \to \quad \mathcal{E} = R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{1}{C}q \quad \to \quad q(t) = C\mathcal{E}\left(1 - e^{-\frac{t}{RC}}\right).$$

Similarly, we can find the current at any time t by taking the time derivative of $q(t)$:

$$i(t) = \frac{\mathrm{d}}{\mathrm{d}t}q(t) = \frac{\mathcal{E}}{R}e^{-\frac{t}{RC}}.$$

Since the potential difference across the capacitor is $\Delta V_C = q/C$, we can also find ΔV_C at any time t:

$$\Delta V_C(t) = \frac{q(t)}{C} = \mathcal{E}\left(1 - e^{-\frac{t}{RC}}\right).$$

The quantity $\tau = RC$ is called the **RC time constant** of the circuit.

- After a time $t = \tau$, the charge on the capacitor will have reached about 63% of its final value, and the current will have dropped to about 37% of its initial value.
- Using the time constant, we can rewrite our equations:

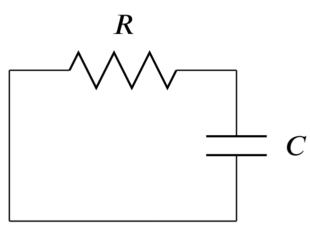
$$q(t) = C\mathcal{E}\left(1 - e^{-t/\tau}\right) \qquad i(t) = \frac{\mathcal{E}}{R}e^{-t/\tau} \qquad \Delta V_C(t) = \mathcal{E}\left(1 - e^{-t/\tau}\right)$$

Note the following limits:

- $\lim_{t\to\infty} q(t) = C\mathcal{E}$: After a very long time, the capacitor approaches its maximum charge of $q_{\max} = C\mathcal{E}$.
- $i(0) = \mathcal{E}/R$ and $\lim_{t\to\infty} i(t) = 0$: The current starts at its maximum value of $i_{\max} = \mathcal{E}/R$ and eventually drops to zero.

- $\lim_{t\to\infty} \Delta V_C(t) = \mathcal{E}$: After a very long time, the voltage across the capacitor approaches its maximum value of $\Delta V_{\text{C,max}} = \mathcal{E}$.

Discharging an RC Series Circuit



Now, let's assume that we have an RC-series circuit with a fully charged capacitor, so $q(0) = q_0$. Then, the battery used to charge the capactor is removed, and now the capacitor is allowed to discharge through the resistor. We call this **discharging the capacitor**.

By Kirchhoff's Loop Rule, we can write an equation for the circuit:

$$\Delta V_R + \Delta V_C = 0.$$

Substituting $\Delta V_R = IR$ and $\Delta V_C = q/C$ into the equation, we get:

$$R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{1}{C}q = 0 \quad \rightarrow \quad q(t) = q_0 e^{-\frac{t}{RC}}.$$

If we assume we started from a fully charged capacitor via emf $\mathcal{E}$, then $q_0 = C\mathcal{E}$, so we can rewrite as:

$$q(t) = C\mathcal{E}e^{-\frac{t}{RC}}.$$

Using $i = \mathrm{d}q/\mathrm{d}t$ and $\Delta V_C = q/C$, we can find the current and voltage across the capacitor at any time t:

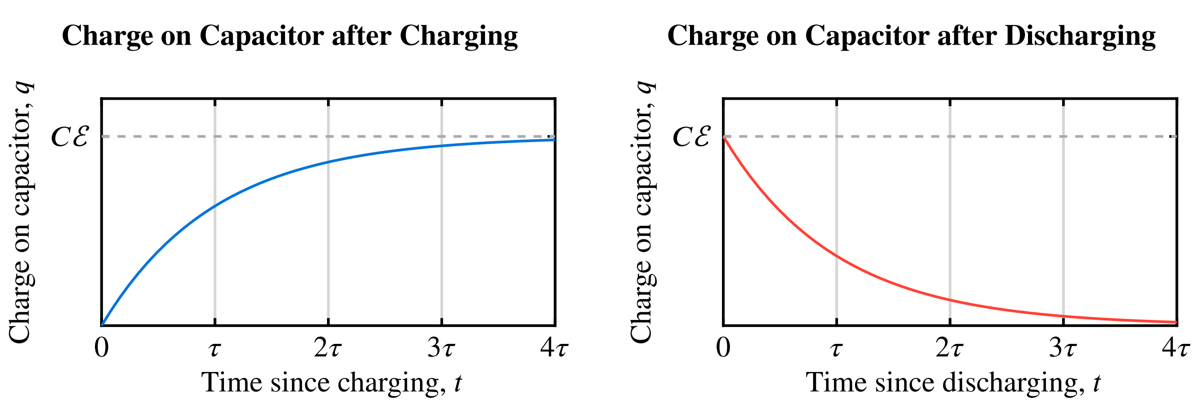
$$i(t) = \frac{\mathrm{d}}{\mathrm{d}t}q(t) = -\frac{\mathcal{E}}{R}e^{-\frac{t}{RC}} \qquad \Delta V_C(t) = \frac{q(t)}{C} = \mathcal{E}e^{-\frac{t}{RC}}.$$

Using the time constant $\tau = RC$, we can rewrite our equations:

$$q(t) = C\mathcal{E}e^{-t/\tau} \qquad i(t) = -\frac{\mathcal{E}}{R}e^{-t/\tau} \qquad \Delta V_C(t) = \mathcal{E}e^{-t/\tau}$$

Realize that:

- $\lim_{t\to\infty} q(t) = 0$: After a very long time, the capacitor will be fully discharged and have zero charge.
- $i_{\text{discharging}}(t) = -i_{\text{charging}}(t)$: The current during discharging is the same as the current during charging, but in the opposite direction. It still decays to zero over time.
- $\lim_{t\to\infty} \Delta V_C(t) = 0$: After a very long time, the voltage across the capacitor will drop to zero.



4 Magnetic Fields and Electromagnetism

4.1 Introduction to Magnetism

- Magnetism is a fundamental force of nature that arises from the motion of electric charges.
- Magnetic fields are produced by moving charges (currents) and exert forces on other moving charges.
  - Magnetic fields are a consequence of the special theory of relativity, and magnetic fields are really just electric fields observed from different reference frames.
- There are three types of magnetic materials (see Types of Magnetic Materials):
  - **ferromagnetic** materials (e.g. iron) are *strongly* attracted to magnetic fields and can be permanently magnetized. For this reason, they are often referred to as *"permanent magnets."*
  - **paramagnetic** materials are weakly attracted by magnetic fields
  - **diamagnetic** materials are weakly repelled by magnetic fields.
- A **magnet** is an object that produces a magnetic field. All magnets (that we know of) have two poles: a **north pole** and a **south pole**.
  - Like poles repel each other, while opposite poles attract each other.
  - The magnetic field lines of a magnet form **closed loops** from the north pole to the south pole.
- The Earth itself is a magnet. Technically, the Earth's North pole is a magnetic south pole, for the north end of a compass needle is attracted to it.
  - The Earth's magnetic field is generated by the motion of molten iron in its outer core, which creates electric currents that produce the magnetic field.
  - Earth's magnetic field is extremely weak, around 50 μT.

4.2 Magnetic Fields and Magnetic Force

- Magnetic fields are produced by moving charges (currents). They exert forces on moving charges.
  - *Moving charges create magnetic fields. Magnetic fields exert forces on moving charges. This is the fundamental principle of electromagnetism.*
- The magnetic field is represented by the vector $\vec{B}$, measured in *Teslas* (T).
  - A 1 Tesla magnetic field exerts a force of 1 Newton on a 1 Coulomb point charge moving at 1 m/s **perpendicular** to the field.
    $\mathrm{T} = \mathrm{N\,s\,C^{-1}\,m^{-1}} = \mathrm{kg\,s^{-2}\,A^{-1}}$ in SI base units.
- The direction of the magnetic field is the direction in which the north pole of a compass needle would point if placed in the field.
- **Magnetic field lines** are a visual representation of the magnetic field.
  - The direction of the magnetic field at any point is tangent to the magnetic field line at that point
  - Stronger magnetic fields have more closely spaced field lines.
  - *Unlike* electric field lines, magnetic field lines form *closed loops*!
    - *Outside* of a bar magnet, magnetic field lines point from the north pole to the south pole.
    - *Inside* of a bar magnet, magnetic field lines point from the south pole to the north pole.

Magnetic Force

Let q be a charge moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$.
The **magnetic force** $\vec{F}_b$ on the charge is:

$$\vec{F}_b = q\vec{v} \times \vec{B}.$$

- The magnetic force is always perpendicular to both the velocity of the charge and the magnetic field.
- The magnetic force does no work on the charge, since it is always perpendicular to the velocity of the charge.
- We can also calculate the magnetic force using current density $\vec{J}$:

$$\mathrm{d}\vec{F}_b = \left(\vec{J} \times \vec{B}\right) \mathrm{d}V \quad \to \quad \vec{F}_b = \iiint_E \left(\vec{J} \times \vec{B}\right) \mathrm{d}V.$$

Lorentz Force Law

The **Lorentz force** $\vec{F}$ on a charge q moving with velocity $\vec{v}$ in both electric and magnetic fields $\vec{E}$ and $\vec{B}$ is:

$$\vec{F} = \vec{F}_e + \vec{F}_b = q\vec{E} + q\vec{v} \times \vec{B} = q\left(\vec{E} + \vec{v} \times \vec{B}\right).$$

The Lorentz force describes how charged particles behave in electromagnetic fields.

Magnetic Force on Current-Carrying Wires

- If charge is moving in a straight line (e.g. a straight wire), then the velocity is constant and we can write the magnetic force as:

$$\vec{F}_b = \frac{q\vec{L}}{t} \times \vec{B} = I\vec{L} \times \vec{B}.$$

  - Here, $\vec{L}$ is the length of a wire segment, and if it takes the charge q a time t to pass through the wire segment, then the current $I = q/t$.
  - The magnetic force on a charge moving through a wire segment is proportional to the length of that segment, the current through that segment, and the magnetic field strength.
  - For a wire that isn't straight, we can divide the wire into tiny segments of length $\mathrm{d}\vec{\ell}$ to get:

$$\mathrm{d}\vec{F}_b = I\,\mathrm{d}\vec{\ell} \times \vec{B} \quad \rightarrow \quad \vec{F}_b = \int_C I\,\mathrm{d}\vec{\ell} \times \vec{B}.$$

- A **coil** has no net magnetic force on it, since the magnetic forces on opposite sides of the coil cancel each other out ($\oint I\,\mathrm{d}\vec{\ell} \times \vec{B} = 0$). However, a coil can still *produce* a magnetic field.
- A **solenoid** is a long coil of wire (a wire formed from multiple coils). The magnetic field inside a solenoid is approximately uniform and parallel to the axis of the solenoid, while the magnetic field outside a solenoid is weak and non-uniform.

4.3 Magnetic Dipoles

- *Recall:* an electric dipole consists of two equal and opposite charges separated by a small distance.
- Unlike charges which can exist independently, magnetic poles always come in pairs[3] (north and south). Thus, all magnets are **magnetic dipoles**.
- The magnetic field of a magnetic dipole is similar to the electric field of an electric dipole, but with some differences:
  - The magnetic field lines of a magnetic dipole form closed loops, while the electric field lines of an electric dipole start at the positive charge and end at the negative charge.
  - The magnetic field of a magnetic dipole decreases with distance by the cube ($B \propto 1/r^3$), while the electric field of an electric dipole decreases with distance by the square ($B \propto 1/r^2$).
- The **magnetic dipole moment** $\vec{\mu}$ of a magnetic dipole is a vector that points from the south pole to the north pole.

  In a region enclosed by a current-carrying coil with area A, the magnetic dipole moment is:

  $$\vec{\mu} = IA\hat{n} = I\underbrace{\vec{A}}_{A\hat{n}}.$$

  where:
  - I is the current through the coil
  - A is the area of the coil
  - $\hat{n}$ is the unit vector perpendicular to the plane of the coil, pointing in the direction given by the right-hand rule (curl your hands in the direction of current, and your thumb will point in $\hat{n}$)

  For a solenoid with N loops, the magnetic dipole moment is:

  $$\vec{\mu} = NI\vec{A}.$$

- A magnetic field exerts a torque $\vec{\tau}$ on a magnetic dipole:

  $$\vec{\tau} = \vec{\mu} \times \vec{B}.$$

  The magnetic dipole will experience a torque that tends to align it with the magnetic field.

  This means that there is the potential for work to be done to align a magnetic dipole with a magnetic field, given by the potential energy U_μ of the dipole:

  $$U_\mu = -\vec{\mu} \cdot \vec{B}.$$

In summary,
- if we place a current-carrying coil in a magnetic field, it will experience a torque that tends to align it with the field.
- the potential energy associated with this dipole is the work required to align the dipole with the field.

[3] Although we aren't able to *disprove* magnetic monopoles, none have been discovered.

4.4 Magnetic Fields from Moving Charges

Biot-Savart Law

Whereas the formulas above allow us to find the magnetic force on a moving charge in an *existing* magnetic field (the first part of electromagnetism), moving charges also create their *own* magnetic fields (the second part of electromagnetism).

The **Biot-Savart Law** allows us to calculate the magnetic field *created* by a moving charge or current.

Biot-Savart Law

Let I be the current through a small wire segment $\mathrm{d}\vec{\ell}$. The magnetic field $\mathrm{d}\vec{B}$ at a point P due to this wire segment is:

$$\mathrm{d}\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\vec{\ell}\times\hat{r}}{r^2}$$

where:
- $\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{N\,A^{-2}}$ is the **permeability of free space**
- $\hat{r}$ is the unit vector from the wire segment to point P
- r is the distance from the wire segment to point P

If a wire is curved in a path C, then the total magnetic field at point P is:

$$\vec{B} = \frac{\mu_0}{4\pi}\int_C \frac{I\,\mathrm{d}\vec{\ell}\times\hat{r}}{r^2}.$$

For point charge q with velocity $\vec{v}$, the Biot-Savart Law can be simplified as:

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}.$$

By the definition of the cross product, the magnitude of the magnetic field contribution from the wire segment is:

$$\left|\mathrm{d}\vec{B}\right| = \mathrm{d}B = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\ell\sin\theta}{r^2}$$

where θ is the angle between $\mathrm{d}\vec{\ell}$ and $\hat{r}$.

Example: Magnetic field from a straight current-carrying wire

An infinitely long, straight wire carries a current I. What is the magnetic field $\vec{B}$ at a point P a distance h away from the wire?

By the Biot-Savart Law, the magnetic field at point P is:

$$\vec{B} = \frac{\mu_0}{4\pi} \int_{\text{wire}} \frac{I \, \mathrm{d}\vec{\ell} \times \hat{r}}{r^2}.$$

Say that the wire lies along the x-axis, and point P is at coordinates $(0, h)$. Along the wire, I is constant, so:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int_{\text{wire}} \frac{\mathrm{d}\vec{\ell} \times \hat{r}}{r^2}.$$

Evaluate $\mathrm{d}\vec{\ell} \times \hat{r}$ using the definition of the cross product:

$$\mathrm{d}\vec{\ell} \times \hat{r} = \mathrm{d}\ell \sin\theta \, \hat{n}.$$

Notice that $\hat{n}$ is perpendicular to the plane formed by $\mathrm{d}\vec{\ell}$ and $\hat{r}$, so it is constant. However, $\sin\theta$ is *not* constant. Together, we get:

$$\vec{B} = \frac{\mu_0 I \hat{n}}{4\pi} \int_{-\infty}^{+\infty} \frac{\mathrm{d}x \sin\theta}{r^2}.$$

Using trigonometry, $\sin\theta = h/r$ and $r^2 = x^2 + h^2$. Thus,

$$\begin{aligned} \vec{B} &= \frac{\mu_0 I \hat{n}}{4\pi} \int_{-\infty}^{+\infty} \frac{\mathrm{d}x \, h}{\sqrt{x^2 + h^2}\left(x^2 + h^2\right)} \\ &= \frac{\mu_0 I h \hat{n}}{4\pi} \int_{-\infty}^{+\infty} \frac{\mathrm{d}x}{\left(x^2 + h^2\right)^{3/2}}. \end{aligned}$$

Evaluating the integral gives:

$$\vec{B} = \frac{\mu_0 I h \hat{n}}{4\pi} \left(\frac{2}{h^2}\right) = \frac{\mu_0 I}{2\pi h} \hat{n}.$$

Ampere's Law

The Biot-Savart Law allows us to calculate the magnetic field created by a current, but it can be difficult to apply in practice.

Ampere's Law provides an alternative way to calculate magnetic fields, especially for cases with high symmetry.

Ampere's Law

Let C be a closed, often imaginary loop, called an **Amperian loop**. Let I_{enclosed} be the net current passing through any surface enclosed[4] by C. **Ampere's Law** states that:

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{s} = \mu_0 I_{\text{enclosed}}.$$

We often call the quantity $\oint_C \vec{B} \cdot \mathrm{d}\vec{s}$ the **magnetic circulation** over C. Thus, Ampere's Law states that the magnetic circulation around a closed loop is proportional to the net current passing through the area enclosed by that loop.

Example: Magnetic field from a straight current-carrying wire, revisited

A long, straight wire with radius R carries a current I. What is the magnetic field $\vec{B}$ at a point P a distance r away from the center of the wire?

Create a circular Amperian loop of radius $r > R$ centered on the wire. By symmetry, the magnetic field $\vec{B}$ has the same magnitude at every point on the loop and is tangent to the loop, so B is constant.

Thus, by Ampere's Law:

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{s} = B \oint_C \mathrm{d}s = B(2\pi r) = \mu_0 I.$$

Solving for B gives:

$$B = \frac{\mu_0 I}{2\pi r} \text{ for } r > R.$$

This is the same result we would get from the Biot-Savart Law, but Ampere's Law is much easier to apply in this case due to the symmetry of the problem.

Ampere's Law also allows us to find the magnetic field inside the wire (where $r < R$), which is:

$$B = \frac{\mu_0 I r}{2\pi R^2} \text{ for } r < R.$$

[4] by "enclosed", we mean that the surface has a boundary defined by C. That is, if S is the surface, $\partial S \equiv C$.

Magnetic Fields of Solenoids

- A solenoid is a wire formed from multiple coils, called **turns**.
  - The magnetic field inside a solenoid is approximately uniform and parallel to the axis of the solenoid
  - The magnetic field outside a solenoid is weak and non-uniform.
- The magnetic field *inside* a solenoid with current I can be calculated using Ampere's Law:
  - Consider a rectangular Amperian loop that extends inside and outside the solenoid.
  - The magnetic field outside the solenoid is negligible, so the contribution to the line integral from the outside part of the loop is approximately zero.
  - The magnetic field inside the solenoid is approximately uniform, so the contribution to the line integral from the inside part of the loop is approximately BL, where L is the length of the section of the Amperian loop that is inside the solenoid.
  - Along the sides, the magnetic field is perpendicular to the path of integration, so the contribution to the line integral from the sides is zero.
  - If our loop encloses N turns of the solenoid, then the total current enclosed by the loop is NI.

  Thus, by Ampere's Law:

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{\ell} = BL = \mu_0 N I \quad \to \quad B = \frac{\mu_0 N I}{L}.$$

- We call $n = N/L$ the **turn density** of the solenoid, which is the number of turns per unit length. This allows us to rewrite the formula for the magnetic field inside a solenoid as:

$$B = \mu_0 n I.$$

Differential Form of Ampere's Law

Using Stokes' Theorem and the integral definition of current density, we can rewrite Ampere's Law in differential form:

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{\ell} = \mu_0 I_{\text{enclosed}}$$

$$\iint_S \left(\vec{\nabla} \times \vec{B}\right) \cdot \mathrm{d}\vec{A} = \mu_0 \iint_S \vec{J} \cdot \mathrm{d}\vec{A}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}.$$

Ampere's Law (Differential Form)

Let a current density $\vec{J}$ produce a magnetic field $\vec{B}$. **Ampere's Law** in differential form states that:

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}.$$

That is, the *curl* of the magnetic field is proportional to the current density that produces it.

Extension: Magnetic Vector Potential

Recall: the electric field $\vec{E}$ can be expressed using the gradient of the *electric potential* V: $\vec{E} = -\vec{\nabla}V$.

Similarly, the magnetic field $\vec{B}$ can be expressed as the curl of a vector field called the **magnetic vector potential** $\vec{A}$:

$$\vec{B} = \vec{\nabla} \times \vec{A}.$$

Recall that the divergence of the magnetc field is zero ($\vec{\nabla} \cdot \vec{B} = 0$, Gauss' Law for Magnetism), which means that the magnetic field is guaranteed to be the curl of some vector field.

Magnetic Vector Potential

The **magnetic vector potential** $\vec{A}$ is a vector field whose curl gives the magnetic field $\vec{B}$:

$$\vec{B} = \vec{\nabla} \times \vec{A}.$$

- The magnetic vector potential is a mathematical tool that allows us to calculate the magnetic field in certain situations, especially when dealing with complex geometries or time-varying fields.
- The magnetic vector potential has units of T m ≡ Wb/m.
- The magnetic vector potential is *not* unique, since we can add the gradient of any scalar field to $\vec{A}$ without changing the magnetic field (since the curl of a gradient is zero)

Biot-Savart Law (Vector Potential Form)

Let I be the current through a wire segment $\mathrm{d}\vec{\ell}$. The magnetic vector potential $\mathrm{d}\vec{A}$ at a point P due to this wire segment is:

$$\mathrm{d}\vec{A} = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\vec{\ell}}{r},$$

where r is the distance from the wire segment to point P.

Ampere's Law in magnetic vector potential form can be derived by substituting $\vec{B} = \vec{\nabla} \times \vec{A}$:

$$\vec{\nabla} \times \left(\vec{\nabla} \times \vec{A}\right) = \mu_0 \vec{J}$$

$$\cancel{\vec{\nabla}\left(\vec{\nabla} \cdot \vec{A}\right)} - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}.$$

Also, by Stokes' Theorem, if C is a closed loop and S is any surface with boundary C, then:

$$\oint_C \vec{A} \cdot \mathrm{d}\vec{\ell} = \iint_S \left(\vec{\nabla} \times \vec{A}\right) \cdot \mathrm{d}\vec{A} = \iint_S \vec{B} \cdot \mathrm{d}\vec{A} = \Phi_B.$$

The circulation of the magnetic vector potential around a closed loop is equal to the magnetic flux Φ_B through any surface bounded by that loop.

4.5 Magnetic Properties of Materials

Magnetization and the H-field

At the microscopic level, electrons have a property called **spin** that gives them an intrinsic magnetic dipole moment.

- Most electrons are paired up with another electron of opposite spin, so their magnetic dipole moments cancel each other out.
- Some materials have unpaired electrons, resulting in a *net dipole moment*, which can interact (i.e., align) with external magnetic fields.

The density of these microscopic dipoles is called the **magnetization** of the material:

Magnetization

The **magnetization** or **magnetic polarization** $\overrightarrow{M}$ is the net magnetic dipole moment per unit volume:

$$\overrightarrow{M} = \frac{\mathrm{d}\vec{\mu}}{\mathrm{d}V} \quad \to \quad \vec{\mu} = \iiint_V \overrightarrow{M}\,\mathrm{d}V.$$

For this reason, magnetization is sometimes called *magnetic moment density*.

- An unmagnetized object has $\overrightarrow{M} = \vec{0}$ everywhere (i.e., dipoles cancel)
- Magnetization is a vector quantity, measured in units of A/m.

While $\overrightarrow{M}$ is the internal or *bound* magnetization of the material, an external or *applied* magnetic field created by other currents can realign the dipoles. The applied magnetic field is called the ***H*-field** or **magnetic field intensity**, denoted $\overrightarrow{H}$:

Magnetic Field Intensity (H-field)

The net magnetic field $\vec{B}$ inside a material is proportional to the sum of the applied field $\overrightarrow{H}$ and the field created by the magnetization of the material $\overrightarrow{M}$:

$$\vec{B} = \mu_0\left(\overrightarrow{H} + \overrightarrow{M}\right).$$

The constant of proportionality μ_0 is the permeability of free space.

- In the above equation, we are strictly looking at the magnetic field *inside* the material.
- The H-field is measured in units of A/m, the same as magnetization.
- The H-field is the magnetic field that would exist in the absence of any magnetization. It is the field that we apply to a material to induce magnetization.

Linear Magnetic Materials and Magnetic Permeability

A **linear magnetic material** is one where its magnetization is proportional to the H-field:

$$\overrightarrow{M} = \chi \overrightarrow{H} \text{ for a linear magnetic material,}$$

where the constant of proportionality χ is called the **magnetic susceptibility** of the material. It is a dimensionless quantity that measures how much a material becomes magnetized in response to an applied magnetic field. Most materials are approximately[5] linear.

- For certain materials, χ can be negative! This means that the material becomes magnetized in the *opposite* direction of the applied field (see diamagnetism).
- If $|\chi|$ is high, then the material is easily able to form internal magnetic fields in response to an external magnetic field.

Since $\overrightarrow{M} = \chi \overrightarrow{H}$, then for a linear magnetic material:

$$\vec{B} = \mu_0\left(\overrightarrow{H} + \overrightarrow{M}\right) = \mu_0\left(\overrightarrow{H} + \chi\overrightarrow{H}\right) = \mu_0(1+\chi)\overrightarrow{H}.$$

The **absolute magnetic permeability** μ is the constant of proportionality between $\vec{B}$ and $\overrightarrow{H}$:

$$\vec{B} = \mu \overrightarrow{H}.$$

Magnetic permeability measures how strongly a material *supports* an applied magnetic field. It is measured in units of T m/A. The **relative magnetic permeability** μ_r of a material is the ratio of its absolute permeability to the permeability of free space:

$$\mu_r \overset{\text{def}}{=} \frac{\mu}{\mu_0} = \frac{\mu_0(1+\chi)}{\mu_0} = 1 + \chi.$$

κ or κ_m can be used to denote μ_r (similar to how either ε_r or κ can be used for relative permittivity).

Types of Magnetic Materials

Magnetic materials are generally classified in terms of their magnetic susceptibility χ:

- **Paramagnetic** materials such as aluminum have a small positive magnetic susceptibility ($\chi > 0$), meaning they are weakly attracted to magnetic fields.
- **Diamagnetic** materials such as copper have a small negative magnetic susceptibility ($\chi < 0$), meaning they are weakly repelled by magnetic fields.
- **Ferromagnetic** materials such as iron have a large positive magnetic susceptibility ($\chi \gg 0$), meaning they are strongly attracted to magnetic fields and can retain magnetization even after the external field is removed (*"permanent magnets"*).
  - Ferromagnetic materials are made up of small regions called **magnetic domains**, where the magnetic moments of atoms are aligned in the same direction. When an external magnetic field is applied, these domains can grow and align with the field, leading to a strong overall magnetization.

[5] Most materials are approximately linear for sufficiently small H-fields, but some materials (e.g. ferromagnets) can become noticeably nonlinear at higher field strengths.

5 Electromagnetic Induction

5.1 Magnetic Flux and Faraday’s Law

Magnetic Flux

Magnetic Flux

The **magnetic flux** Φ_B through a surface S is defined as:

$$\Phi_B = \iint_S \vec{B} \cdot \mathrm{d}\vec{A}.$$

- The magnetic flux through a surface is a measure of the amount of magnetic field passing through that surface.
- The units of magnetic flux are Webers (Wb), where 1 Wb = 1 T m^2.

Gauss' Law for Magnetism

The magnetic flux through a closed surface is always zero, since magnetic field lines form closed loops. This is known as **Gauss’s Law for Magnetism**:

$$\underset{\substack{\text{closed}\\\text{surface}}}{\Phi_B} = \oiint_S \vec{B} \cdot \mathrm{d}\vec{A} = 0 \quad \to \quad \vec{\nabla} \cdot \vec{B} = 0 \text{ by the Divergence Theorem.}$$

This is one of Maxwell’s Equations.

Faraday’s Law of Induction

- When the magnetic flux through a loop of current changes, current begins flowing in the loop. This phenomenon is called **electromagnetic induction**, and the current that flows is **induced current**.
- The effective electromotive force $\mathcal{E}$ that would have caused the induced current is called the **induced electromotive force** (or just *induced emf*).
    - That is, if the induced current is I and the resistance of the loop is R, then $\mathcal{E} = IR$.
    - If the loop has N turns, then the total induced emf is the sum of the emf from each turn: $\mathcal{E} = NIR$.
- Michael Faraday that this induced emf is proportional to the rate of change of magnetic flux through the loop. This is known as **Faraday’s Law of Induction**:

$$|\mathcal{E}| = \left|\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}\right|.$$

If our loop has N turns, then we must consider each of the turns:

$$|\mathcal{E}| = N\left|\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}\right|.$$

- The induced current always flows in a direction that opposes the change in magnetic flux. That is, the induced current will create its own magnetic field to oppose the change. This is known as **Lenz's Law**.
  - For example, if the magnetic flux through a loop is increasing, then the induced current will create a magnetic field that would decrease the flux.
  - If the magnetic flux through a loop is decreasing, then the induced current will create a magnetic field that would increase the flux.
  - A negative sign is used in Faraday's Law of Induction to indicate this opposition:

$$\mathcal{E} = -N\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}.$$

Faraday's Law of Induction

Let a coil with N turns be placed in a magnetic field such that its flux through the coil is $\Phi_B(t)$. Then the *induced emf* $\mathcal{E}$ in the coil is given by **Faraday's Law of Induction**:

$$\mathcal{E} = -N\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}.$$

The negative sign is described by **Lenz's Law**.

Induced Electric Fields

If an emf is induced in a circuit, then there is work being done on charges in the circuit. Since magnetic forces do no work on charges, the work must be done by an electric field. Thus, there must be an **induced electric field** in the circuit that causes the induced current to flow.

The work done by the induced electric field to move a unit charge q around the coil is:

$$W = \oint_{\text{coil}} q\vec{E}\cdot \mathrm{d}\vec{\ell}.$$

By the definition of emf:

$$\mathcal{E} = \frac{W}{q} = \oint_{\text{coil}} \vec{E}\cdot \mathrm{d}\vec{\ell}.$$

This allows us to rewrite Faraday's Law as:

$$\oint_{\text{coil}} \vec{E}\cdot \mathrm{d}\vec{\ell} = -N\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}.$$

For a coil with $N = 1$ turn, we can expand the magnetic flux as follows:

$$\oint_{\text{coil}} \vec{E}\cdot \mathrm{d}\vec{\ell} = -\frac{\mathrm{d}}{\mathrm{d}t}\iint_S \vec{B}\cdot \mathrm{d}\vec{A}.$$

where S is the surface enclosed by the coil.

In a general sense, $\vec{B}$ and consequently Φ_B can vary not only with time but also with position, so it is more accurate to use a partial time derivative instead:

$$\oint_{\text{coil}} \vec{E} \cdot \mathrm{d}\vec{\ell} = -\frac{\partial}{\partial t} \iint_S \vec{B} \cdot \mathrm{d}\vec{A}.$$

This is the **integral form** of Faraday's Law of Induction, which is one of Maxwell's Equations. We can also write it in **differential form** using Stokes' Theorem:

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}.$$

Faraday's Law of Induction (Integral and Differential Forms)

Let C be a closed curve representing a loop of current, and let S be a surface enclosed by C. Then Faraday's Law of Induction can be expressed in two equivalent forms:

$$\oint_C \vec{E} \cdot \mathrm{d}\vec{\ell} = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{A}. \qquad (\textbf{integral form})$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}. \qquad (\textbf{differential form})$$

Eddy Currents

So far we have considered conductors shaped like wires, where induced currents are confined to a one-dimensional path. But if a *bulk* (extended) conductor moves through a non-uniform magnetic field – or sits in a time-varying one – the induced emfs drive closed loops of current *within the body of the conductor itself.* These are called **eddy currents**.

By Lenz's Law, eddy currents flow in whichever direction opposes the change in flux that produced them.

In other words, *a changing magnetic flux through a bulk conductor induces eddy currents, which are small, isolated loops of current that create magnetic fields to oppose the change in flux.*

Electric Motors and Back Emf

- An **electric motor** is a device that converts electrical energy into mechanical energy. It consists of a current-carrying coil placed in a magnetic field. The magnetic field exerts a torque on the coil, causing it to rotate.
- As the coil rotates, the magnetic flux through it changes, and Faraday's Law tells us an emf must be induced. By Lenz's Law, this induced emf opposes the applied voltage driving the motor. This is called the **back emf**, $\mathcal{E}_{\text{back}}$.
- Applying Kirchhoff's voltage law around the motor circuit:

$$V - \mathcal{E}_{\text{back}} = IR \quad \rightarrow \quad I = \frac{V - \mathcal{E}_{\text{back}}}{R}.$$

5.2 Motional Electromotive Force

When a conductor moves through a magnetic field[6], the magnetic flux through the conductor can change, which induces an emf in the conductor. This is known as **motional emf**.

In this way, a potential difference is created across the conductor, allowing us to treat the conductor as a circuit element with an emf. *By moving a conductor through a magnetic field, we convert mechanical energy into electrical energy!*

Recall that the fundamental definition of emf is the work done per unit charge to move a charge around a closed loop, $\mathcal{E} = W/q$. In the case of motional emf, the work is done by the *magnetic force*[7] on the conductor as it moves through the magnetic field:

$$\mathcal{E} = \frac{W}{q} = \frac{1}{q}\oint_{\text{loop}} \vec{F}_b \cdot \mathrm{d}\vec{\ell} = \frac{1}{q} = \oint_{\text{loop}} \left(q\vec{v} \times \vec{B}\right) \cdot \mathrm{d}\vec{\ell} = \oint_{\text{loop}} \left(\vec{v} \times \vec{B}\right) \cdot \mathrm{d}\vec{\ell}.$$

Motional Emf

When a conductor along a path C moves through a magnetic field $\vec{B}$ at velocity $\vec{v}_{\text{conductor}}$, the **motional emf** $\mathcal{E}$ induced in the conductor is given by:

$$\mathcal{E} = \oint_C \left(\vec{v}_{\text{conductor}} \times \vec{B}\right) \cdot \mathrm{d}\vec{\ell}.$$

For a straight conductor of length ℓ moving perpendicularly through a uniform magnetic field[8]:

$$\mathcal{E} = B\ell v_{\text{conductor}}.$$

Deriving Motional Emf using Faraday's Law

Let a conductor of length ℓ move with velocity $\vec{v} = v\hat{\imath}$ perpendicularly through a uniform magnetic field $\vec{B} = B\hat{k}$. Its ends form a closed loop with a stationary conductor, forming a circuit whose area is increasing as the conductor moves.

Let the moving conductor be a distance x from the left side of the loop. Then the area of the loop is $A = \ell x$, and the magnetic flux through the loop is $\Phi_B = BA = B\ell x$. The rate of change of magnetic flux is:

$$\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(BA) = B\frac{\mathrm{d}A}{\mathrm{d}t} = B\ell\frac{\mathrm{d}x}{\mathrm{d}t} = B\ell v.$$

Thus, by Faraday's Law, the induced emf is:

$$\mathcal{E} = -B\ell v.$$

We call this the **motional emf** since it is caused by the *motion* of the conductor through a magnetic field.

[6] or, the magnetic field can be moving relative to the conductor.

[7] It is true that magnetic force can *never* do work on moving charges, but in this case the conductor itself is moving. That is, the magnetic force does mechanical work on the conductor, and the energy gained from that work is converted into electrical energy (the actual EMF).

[8] The conductor of length ℓ should be connected to another conductor to form a closed loop which is not influenced by the magnetic field.

Realize that as the conductor moves, it feels a magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ that pushes charges in the conductor along the direction of $\hat{\jmath}$. This is the induced current that creates the motional emf.

Example: Square Coil Falling into a Magnetic Field

A square coil with mass m, side length s, and resistance R falls in a gravitational field $-g\hat{\jmath}$ into a region with a uniform magnetic field $B\hat{k}$. As the square coil is entering the magnetic field, what is its terminal speed v?

As the coil enters the magnetic field, the magnetic flux through the loop increases. Let y be the distance the bottom edge of the coil has penetrated into the field, so the area of the coil within the magnetic field is $A = sy$.

Thus, the magnetic flux is:

$$\Phi_B = BA = Bsy.$$

By Faraday's Law:

$$|\mathcal{E}| = \frac{\mathrm{d}\Phi_B}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(Bsy) = Bs\frac{\mathrm{d}y}{\mathrm{d}t}.$$

Since $v = \mathrm{d}y/\mathrm{d}t$, we have:

$$\mathcal{E} = Bsv. \qquad \text{(motional emf!)}$$

Using Ohm's Law, the induced current I flowing through the coil is:

$$I = \frac{\mathcal{E}}{R} = \frac{Bsv}{R}.$$

Only the bottom side of the coil contributes to its net magnetic force, which is given by:

$$\vec{F}_b = I(-s\hat{\imath}) \times B\hat{k} = IsB\hat{\jmath} = \frac{Bsv}{R}sB\hat{\jmath} = \frac{B^2s^2v}{R}\hat{\jmath}.$$

At terminal velocity, the upward magnetic force balances the downward gravitational force $-mg\hat{\jmath}$:

$$\frac{B^2s^2v}{R} = mg \quad \rightarrow \quad v = \frac{mgR}{B^2s^2}.$$

Deriving Motional Emf using Newton's Second Law

We can arrive at the same result without ever invoking Faraday's Law, by looking directly at the forces on the charges inside the moving rod.

Consider a free charge q inside the rod. Because the rod moves with velocity $\vec{v} = v\hat{\imath}$ through $\vec{B} = B\hat{k}$, the charge feels a magnetic force

$$\vec{F}_b = q\vec{v} \times \vec{B},$$

of magnitude qvB, directed along the length of the rod. This force drives positive charges toward one end and negative charges toward the other.

As charge accumulates at the ends, an electric field $\vec{E}$ builds up along the rod, pointing opposite to $\vec{F}_b$. Charges stop accumulating once the electric force balances the magnetic force:

$$qE = qvB \quad \rightarrow \quad E = vB.$$

The potential difference across the rod is then just E integrated along its length:

$$\mathcal{E} = \int_0^\ell E \,\mathrm{d}s = \int_0^\ell vB \,\mathrm{d}s = B\ell v,$$

matching the result from Faraday's Law. The sign simply depends on which end we call positive.

This derivation makes clear where the emf physically comes from: it is the magnetic force on the charge carriers, reinterpreted as an effective electric field along the rod.

Power Dissipated by Motional Emf

To maintain a constant velocity v of the conductor, an external force must be applied to counteract the magnetic force on the charges in the conductor.

As current flows along the $\hat{\jmath}$ direction in the conductor, the magnetic force on the wire is $\vec{F} = I\ell\hat{\jmath} \times B\hat{k} = I\ell B\,\hat{\imath}$. We must apply an external force $\vec{F}_a = -I\ell B\,\hat{\imath}$ to maintain a constant velocity. The power dissipated by the motional emf is:

$$P = \vec{F} \cdot \vec{v} = -I\ell B\,\hat{\imath} \cdot v\,\hat{\imath} = I\ell Bv.$$

Hall Effect

When a current-carrying conductor is placed in a magnetic field perpendicular to the current, the magnetic force $\vec{F}_b = q\vec{v} \times \vec{B}$ pushes charge carriers sideways, causing them to accumulate on one face of the conductor. This charge buildup creates an electric field that opposes further accumulation. At equilibrium the electric and magnetic forces balance, producing a steady **Hall voltage** V_H or **Hall emf** across the width of the conductor.

Hall Voltage

For a conductor of thickness d carrying current I in a magnetic field $\vec{B}$, the **Hall voltage** is:

$$V_H = \frac{IB}{nqd},$$

where n is the charge-carrier number density and q is the carrier charge.

Note

While the Hall emf is not a motional emf, it is derived the same way as the motional emf was derived using Newton's Second Law.

- The sign of V_H reveals whether the carriers are (effectively) positive or negative.
- A **Hall-effect sensor** is a device that uses the Hall effect to measure magnetic field strength. By applying a known current I and measuring the resulting Hall voltage V_H, we can calculate the magnetic field using the formula above:

$$B = \frac{nqdV_H}{I}.$$

5.3 Inductors and Self-Inductance

- An **inductor** is a circuit element that resists changes in current. A common example of an inductor is a solenoid, which stores energy in its magnetic field.
  - While a resistor resists current by dissipating energy as heat, an inductor resists changes in current by storing energy in its magnetic field. This energy can then be released back into the circuit (as a source of EMF) when the current changes again.
- **Inductance** is a property of a circuit element that quantifies how much it resists changes in current. The higher the inductance, the more it resists changes in current.
  - The unit of inductance is the **Henry** ("H"), where 1 H = 1 Wb/A = 1 Ω s.

Self-Inductance

- When a current I flows through a coil, a magnetic field is produced around the wires of the coil. This creates a magnetic flux Φ_B through the coil itself.

  If I changes, so does Φ_B. By Faraday's Law, an emf is induced in the coil that opposes the change in current. This is called **self induction**. *The current in the coil creates a magnetic field that opposes changes in the current itself.*

- The magnetic flux through the coil itself that is generated by a current I flowing through the coil depends on how the coil is shaped.

  The magnetic flux Φ_B created by running a current I through the coil is proportional to that current:

  $$\Phi_B = LI.$$

  The constant of proportionality L is called the **self-inductance** of the coil. It is one form of *inductance*. It depends only on the geometry of the coil and the permeability of any material inside it.

  How well does the magnetic field generated by the current cause magnetic flux through the same coil?

- For a coil with N turns, we want to look at the current *per turn*, so use:

  $$\Phi_B = \frac{LI}{N} \quad \rightarrow \quad L = N\frac{\Phi_B}{I}.$$

Self-Inductance

If running a current I through a coil of N turns creates a magnetic flux Φ_B through the coil, then the **self-inductance** L of the coil is:

$$L = N\frac{\Phi_B}{I}.$$

Then, the **self-induced emf** from an inductor is given by:

$$\mathcal{E} = -N\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} = -N\frac{\mathrm{d}}{\mathrm{d}t}\left(L\frac{I}{N}\right) = -L\frac{\mathrm{d}I}{\mathrm{d}t}.$$

Example: Self-Inductance of a Solenoid

What is the self-inductance of a long solenoid with turn density n and volume Ω?

Let the solenoid have N turns, length ℓ, and cross-sectional area A. Then $n = N/\ell$ and $\Omega = \ell A$.

The magnetic field inside a long solenoid is $B = \mu_0 N I/\ell$. The magnetic flux through each turn of the solenoid is $\Phi_B = BA = \mu_0 N I A/\ell$.

The self-inductance is then:

$$L = N\frac{\Phi_B}{I} = N\frac{\mu_0 N I A/\ell}{I} = \frac{\mu_0 N^2 A}{\ell}.$$

We can rewrite this in terms of the turn density and volume:

$$L = \frac{\mu_0 N^2 A\ell}{\ell^2} = \mu_0 n^2 \Omega.$$

Note

The solenoid formula above assumed free space (μ_0) inside the coil. If the core is filled with a linear magnetic material of permeability $\mu = \mu_r \mu_0$, the field inside becomes $\vec{B} = \mu n I$, and the self-inductance is amplified by the same factor:

$$L = \mu_r \mu_0 n^2 \Omega = \mu_r L_0.$$

Example: Self-Inductance of a Toroid

A **circular toroid** is a solenoid bent into a circle. What is the self-inductance of a toroid with N turns, *mean inner radius* r, and cross-sectional area A?

The magnetic field inside a toroid is $B = \mu_0 N I/(2\pi r)$. The magnetic flux through each turn of the toroid is $\Phi_B = BA = \mu_0 N I A/(2\pi r)$.

The self-inductance is then:

$$L = N\frac{\Phi_B}{I} = N\frac{\mu_0 N I A/(2\pi r)}{I} = \frac{\mu_0 N^2 A}{2\pi r}.$$

Energy in an Inductor

To build up a current I in an inductor from zero, the source must do work against the back emf. The instantaneous power delivered to the inductor is:

$$P = I\,\mathcal{E}_L = L\,I\frac{\mathrm{d}I}{\mathrm{d}t}.$$

Since $|P| = \mathrm{d}U/\,\mathrm{d}t$, $P\,\mathrm{d}t = \mathrm{d}U$, so:

$$\Delta U = \int P\,\mathrm{d}t = \int LI\;\mathrm{d}I = \frac{1}{2}LI^2.$$

Recall the definition of inductance: $L = N\Phi_B/I$. Substituting this into the energy formula gives:

$$U = \frac{1}{2}LI^2 = \frac{1}{2}N\Phi_B I.$$

Power and Energy in an Inductor

The instantaneous power delivered to an inductor is:

$$P = I\,\mathcal{E}_L = L\,I\frac{\mathrm{d}I}{\mathrm{d}t}.$$

The energy stored in an inductor with current I is:

$$U = \frac{1}{2}LI^2 = \frac{1}{2}N\Phi_B I.$$

Energy Density of a Magnetic Field

This energy is stored inside the magnetic field itself. Just like how we derived the energy density of an electric field using a capacitor, we can derive the energy density of a magnetic field by looking at the energy stored in a solenoid.

Take a long solenoid with turn density n and volume Ω. The energy stored in the solenoid is:

$$U = \frac{1}{2}LI^2 = \frac{1}{2}\mu_0 n^2 \Omega I^2.$$

The magnetic field inside the solenoid is $B = \mu_0 nI$, so we can rewrite the energy as:

$$U = \frac{B^2}{2\mu_0}\Omega.$$

The energy density of the magnetic field is then:

$$u_B = \frac{U}{\Omega} = \frac{B^2}{2\mu_0}.$$

5.4 Mutual Inductance

Now consider two nearby coils. A current I_1 in coil 1 produces a magnetic field, some fraction of whose flux Φ_{21} passes through coil 2 (say coil 2 has N_2 turns). Φ_{21} is the magnetic flux through each turn of coil 2 *due to* current in coil 1, "*flux through coil 2 due to coil 1*."

Just as before, the flux passing through coil 2 is proportional to the current:

$$N_2 \Phi_{21} = M_{21} I_1.$$

The constant of proportionality M_{21} is called the **mutual inductance** of coil 2 with respect to coil 1.

The idea behind mutual inductance is that if I_1 changes, an emf is induced in coil 2:

$$\mathcal{E}_2 = -M \frac{\mathrm{d}I_1}{\mathrm{d}t}.$$

Mutual Inductance

Let coil 1 carry a current I_1 that produces a magnetic flux Φ_{21} through coil 2. Then the **mutual inductance** M of coil 2 with respect to coil 1 is defined as:

$$M = N_2 \frac{\Phi_{21}}{I_1}.$$

The **mutually induced emf** in coil 2 due to a changing current I_1 in coil 1 is then:

$$\mathcal{E}_2 = -M \frac{\mathrm{d}I_1}{\mathrm{d}t}.$$

- Mutual inductance is symmetric (this is non-trivial):

$$M_{12} = M_{21} = M.$$

- Both self-inductance and mutual inductance depend only on the geometry of the coils and the permeability of any material inside them.
- The mutual inductance between two coils are related to their self-inductances by:

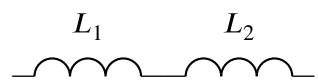
$$M = k\sqrt{L_1 L_2}, \text{where } k \text{ is the } \textbf{coupling coefficient}.$$

  - k ranges from 0 (*no coupling*) to 1 (*perfect coupling*).
  - The coupling coefficient depends on the relative positions and orientations of the coils (e.g. if coils are far apart, k is closer to 0, while if they are close together, k is closer to 1).
- Mutual inductance can cause unwanted interference between inductors in a circuit. Multiple inductors behave ideally when they have a very low coupling coefficient.

Inductors in Series

The potential difference across inductors in series is the sum of the potential differences across each inductor. If there is negligible coupling between then inductors, we can assume that the potential difference across each inductor is $\Delta V_i = L_i \, \mathrm{d}I/\,\mathrm{d}t$:

$$\begin{aligned} \Delta V_{\text{eq}} &= \Delta V_1 + \Delta V_2 + \ldots + \Delta V_n \\ L_{\text{eq}} \frac{\mathrm{d}I}{\mathrm{d}t} &= L_1 \frac{\mathrm{d}I}{\mathrm{d}t} + L_2 \frac{\mathrm{d}I}{\mathrm{d}t} + \ldots + L_n \frac{\mathrm{d}I}{\mathrm{d}t} \\ L_{\text{eq}} &= L_1 + L_2 + \ldots + L_n. \end{aligned}$$



Now, consider two *coupled* inductors with mutual inductance M. The potential difference across each inductor is not just $\Delta V_i = L_i \, \mathrm{d}I/\,\mathrm{d}t$, but also includes the contribution from the other inductor:

$$\begin{aligned} \Delta V_1 &= L_1 \, \mathrm{d}I/\,\mathrm{d}t \pm M \, \mathrm{d}I/\,\mathrm{d}t \\ \Delta V_2 &= L_2 \, \mathrm{d}I/\,\mathrm{d}t \pm M \, \mathrm{d}I/\,\mathrm{d}t. \end{aligned}$$

The orientation of the coils determines the sign of the mutual inductance contribution. If the coils are oriented such that their magnetic fields reinforce each other, then the mutual inductance contribution is positive, and we have:

$$\begin{aligned} \Delta V_{\text{eq}} = L_{\text{eq}} \frac{\mathrm{d}I}{\mathrm{d}t} &= \left(L_1 + L_2 + 2M\right) \frac{\mathrm{d}I}{\mathrm{d}t} \\ L_{\text{eq}} &= L_1 + L_2 + 2M. \end{aligned}$$

That is, the flux created by coil 1 also increases the flux in coil 2.

However, if the coils are oriented such that their magnetic fields oppose each other, then the mutual inductance contribution is negative, and we have:

$$\begin{aligned} \Delta V_{\text{eq}} = L_{\text{eq}} \frac{\mathrm{d}I}{\mathrm{d}t} &= \left(L_1 + L_2 - 2M\right) \frac{\mathrm{d}I}{\mathrm{d}t} \\ L_{\text{eq}} &= L_1 + L_2 - 2M. \end{aligned}$$

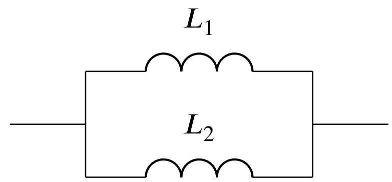
For many inductors in series, we must add or subtract the mutual inductance contributions from each pair of inductors, depending on their orientations:

$$L_{\text{eq}} = L_1 + L_2 + \ldots + L_n \pm 2M_{12} \pm 2M_{13} \pm \ldots$$

Inductors in Parallel

For uncoupled inductors in parallel, the potential difference across each inductor is the same, but the current through each inductor is added together (KJL). Since $\Delta V = L_i\, \mathrm{d}I_i/\mathrm{d}t$, $I_i = \Delta V / L_i$:

$$I_{\mathrm{eq}} = I_1 + I_2 + \ldots + I_n$$
$$\frac{\Delta V}{L_{\mathrm{eq}}} = \frac{\Delta V}{L_1} + \frac{\Delta V}{L_2} + \ldots + \frac{\Delta V}{L_n}$$
$$\frac{1}{L_{\mathrm{eq}}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_n}.$$



Now, consider two *coupled* inductors in parallel with mutual inductance M. The potential difference across each inductor is not just $\Delta V = L_i\, \mathrm{d}I_i/\mathrm{d}t$, but also includes the contribution from the other inductor:

$$\Delta V = L_1\, \mathrm{d}I_1/\mathrm{d}t \pm M\, \mathrm{d}I_2/\mathrm{d}t$$
$$\Delta V = L_2\, \mathrm{d}I_2/\mathrm{d}t \pm M\, \mathrm{d}I_1/\mathrm{d}t.$$

This is a system of two equations with two unknowns (I_1 and I_2), which solves to become:

$$\frac{\mathrm{d}I_1}{\mathrm{d}t} = \frac{L_2 \pm M}{L_1 L_2 - M^2}\Delta V$$
$$\frac{\mathrm{d}I_2}{\mathrm{d}t} = \frac{L_1 \pm M}{L_1 L_2 - M^2}\Delta V.$$

Since $I_{\mathrm{eq}} = I_1 + I_2$, we have:

$$\frac{\mathrm{d}I_{\mathrm{eq}}}{\mathrm{d}t} = \frac{\mathrm{d}I_1}{\mathrm{d}t} + \frac{\mathrm{d}I_2}{\mathrm{d}t} = \frac{L_1 + L_2 \pm 2M}{L_1 L_2 - M^2}\Delta V = \frac{\Delta V}{L_{\mathrm{eq}}}$$
$$\rightarrow \quad L_{\mathrm{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 \pm 2M}.$$

Again, the sign of $2M$ depends on the orientations of the coils. For many inductors in parallel, we would have to set up larger systems. For example, with three inductors in parallel:

$$\Delta V = L_1\, \mathrm{d}I_1/\mathrm{d}t \pm M_{12}\, \mathrm{d}I_2/\mathrm{d}t \pm M_{13}\, \mathrm{d}I_3/\mathrm{d}t$$
$$\Delta V = L_2\, \mathrm{d}I_2/\mathrm{d}t \pm M_{21}\, \mathrm{d}I_1/\mathrm{d}t \pm M_{23}\, \mathrm{d}I_3/\mathrm{d}t$$
$$\Delta V = L_3\, \mathrm{d}I_3/\mathrm{d}t \pm M_{31}\, \mathrm{d}I_1/\mathrm{d}t \pm M_{32}\, \mathrm{d}I_2/\mathrm{d}t.$$

In matrix form, this is:

$$\begin{pmatrix} \Delta V \\ \Delta V \\ \Delta V \end{pmatrix} = \begin{pmatrix} L_1 & \pm M_{12} & \pm M_{13} \\ \pm M_{21} & L_2 & \pm M_{23} \\ \pm M_{31} & \pm M_{32} & L_3 \end{pmatrix} \begin{pmatrix} \mathrm{d}I_1/\mathrm{d}t \\ \mathrm{d}I_2/\mathrm{d}t \\ \mathrm{d}I_3/\mathrm{d}t \end{pmatrix}.$$

After solving for $\mathrm{d}I_1/\mathrm{d}t$, $\mathrm{d}I_2/\mathrm{d}t$, and $\mathrm{d}I_3/\mathrm{d}t$, we can find $\mathrm{d}I_{\mathrm{eq}}/\mathrm{d}t$ and thus L_{eq}.

Energy Stored in Multiple Inductors

The energy stored in multiple inductors is not just the sum of the energies stored in each inductor, but also includes contributions from the mutual inductance between the inductors. For two coupled inductors with mutual inductance M, the total energy stored is:

$$U = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 \pm M I_1 I_2.$$

Extension: Proof that Mutual Inductance is Symmetric

Let the current I_1 flow through loop C_1, and Φ_{21} is the magnetic flux through loop C_2 due to I_1.

The flux through the second coil can be expressed as:

$$\Phi_{21} = \oint_{C_2} \vec{A}_1 \cdot \mathrm{d}\vec{\ell}_2,$$

where $\vec{A}_1$ is the magnetic vector potential created by the current I_1 in coil 1.

By the Biot-Savart Law in vector potential form, the current I_1 in coil 1 creates a vector potential:

$$\vec{A}_1(\vec{r}) = \frac{\mu_0 I_1}{4\pi} \oint_{C_2} \frac{\mathrm{d}\vec{\ell}_1}{|\vec{r} - \vec{r}_1|},$$

where $\vec{r}_1$ is the position vector of points on coil 1. Plugging this into the expression for Φ_{21} gives:

$$\begin{aligned} \Phi_{21} = \oint_{C_2} \vec{A}_1(\vec{r}_2) \cdot \mathrm{d}\vec{\ell}_2 &= \oint_{C_2} \left(\frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{\mathrm{d}\vec{\ell}_1}{|\vec{r}_2 - \vec{r}_1|} \right) \cdot \mathrm{d}\vec{\ell}_2 \\ &= \frac{\mu_0 I_1}{4\pi} \oint_{C_2} \oint_{C_1} \frac{\mathrm{d}\vec{\ell}_2 \cdot \mathrm{d}\vec{\ell}_1}{|\vec{r}_2 - \vec{r}_1|}. \end{aligned}$$

Using definition of mutual inductance, we can find M_{21}:

$$M_{21} = \frac{\Phi_{21}}{I_1} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{\mathrm{d}\vec{\ell}_2 \cdot \mathrm{d}\vec{\ell}_1}{|\vec{r}_2 - \vec{r}_1|}.$$

We can repeat the same process to find Φ_{12} and thus M_{12}:

$$\Phi_{12} = \oint_{C_1} \vec{A}_2 \cdot \mathrm{d}\vec{\ell}_1 = \frac{\mu_0 I_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{\mathrm{d}\vec{\ell}_1 \cdot \mathrm{d}\vec{\ell}_2}{|\vec{r}_1 - \vec{r}_2|},$$

$$M_{12} = \frac{\Phi_{12}}{I_2} = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{\mathrm{d}\vec{\ell}_1 \cdot \mathrm{d}\vec{\ell}_2}{|\vec{r}_1 - \vec{r}_2|}.$$

Since the integrand is symmetric in C_1 and C_2, we have $M_{12} = M_{21}$.

5.5 Circuits with Inductors (LR, LC, LRC)

An inductor produces an emf that opposes changes in current, creating a potential difference across it. The direction of this potential difference depends on whether the current is increasing or decreasing:

- In the direction of current flow, if current is increasing, then the inductor will act like an opposing/*back* emf (voltage drop).
- In the direction of current flow, if current is decreasing, then the inductor will act like a source of emf (voltage gain).

The magnitude of the voltage drop or gain is given by Faraday's Law:

$$\left|\Delta V_L\right| = \left|\mathcal{E}_L\right| = L\left|\frac{\mathrm{d}I}{\mathrm{d}t}\right|.$$

If other inductors in the circuit couple with the inductor, then we also need to include mutual inductances:

$$\left|\Delta V_k\right| = L_k\left|\frac{\mathrm{d}I_k}{\mathrm{d}t}\right| + \sum_{i\neq k} \pm M_{ik}\left|\frac{\mathrm{d}I_i}{\mathrm{d}t}\right|.$$

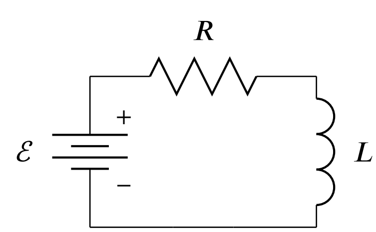
Inductor-Resistor (LR) Series Circuits

An **LR series circuit** is one containing an inductor L and a resistor R in series. Like RC circuits, its behavior is described by a first-order linear ODE. When an EMF source is present, the inductor **energizes** as current builds up. When the EMF source is removed, the inductor **de-energizes** as the current decays.

Energizing an LR Series Circuit

Assume the circuit has been open for a long time, so $i(0) = 0$. At $t = 0$, the switch is closed. Applying Kirchhoff's Loop Rule:

$$\mathcal{E} - iR - L\frac{\mathrm{d}i}{\mathrm{d}t} = 0 \implies L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri = \mathcal{E}.$$



This is a first-order linear ODE whose solution (with $i(0) = 0$) is:

$$i(t) = \frac{\mathcal{E}}{R}\left(1 - e^{-Rt/L}\right).$$

The voltage across the inductor at any time t is:

$$\Delta V_L(t) = L\frac{\mathrm{d}i}{\mathrm{d}t} = \mathcal{E}e^{-Rt/L}.$$

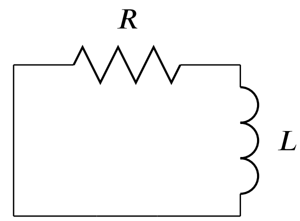
The quantity $\tau = L/R$ is the **LR time constant**, such that the exponential can be simplified to $e^{-t/\tau}$.

Note the following limits:

- $i(0) = 0$: no current flows initially, so the inductor resists any sudden change in current.
- $\lim_{t\to\infty} i(t) = \mathcal{E}/R$: at steady state, the inductor acts as a short circuit, and current is limited only by R.
- $\lim_{t\to\infty} \Delta V_L(t) = 0$: the inductor's back emf decays as the current stabilizes.

De-energizing an LR Series Circuit

Now suppose the battery is removed when the inductor carries current $i(0) = I_0 = \mathcal{E}/R$. By Kirchhoff's Loop Rule (no emf source):



$$-iR - L\frac{\mathrm{d}i}{\mathrm{d}t} = 0 \implies L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri = 0.$$

This separable ODE gives:

$$i(t) = I_0 e^{-t/\tau}.$$

- $i(0) = I_0$: current starts at its maximum value.
- $\lim_{t\to\infty} i(t) = 0$: the energy stored in the inductor is dissipated as heat in R.

LR Series Circuit

For an LR series circuit with emf $\mathcal{E}$, resistance R, and inductance L, the **LR time constant** τ is:

$$\tau = \frac{L}{R}.$$

Energizing (switch closes at $t = 0$, $i(0) = 0$):

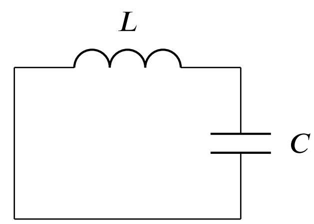
$$i(t) = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right) \qquad \Delta V_L(t) = \mathcal{E}e^{-t/\tau}$$

De-energizing (battery removed at $t = 0$, $i(0) = I_0$):

$$i(t) = I_0 e^{-t/\tau} \qquad \Delta V_L(t) = -I_0 R e^{-t/\tau}$$

Inductor-Capacitor (LC) Series Circuits

An **LC series circuit** contains only an inductor L and capacitor C in series with no resistor and no battery. Suppose the capacitor starts fully charged to $q(0) = Q_0$ and the initial current $i(0) = 0$.



By Kirchhoff's Loop Rule:

$$\Delta V_C - L\frac{\mathrm{d}i}{\mathrm{d}t} = 0 \implies \frac{q}{C} = L\frac{\mathrm{d}i}{\mathrm{d}t}.$$

Since $i = \mathrm{d}q/\,\mathrm{d}t$, this becomes a second-order ODE in q:

$$L\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \frac{1}{C}q = 0 \implies \frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \omega_0^2 q = 0,$$

where $\omega_0 = 1/\sqrt{LC}$ is the **natural angular frequency** of the circuit.

This models **simple harmonic motion** in charge. The characteristic equation $r^2 + \omega_0^2 = 0$ has roots $r = \pm\omega_0 i$, giving the general solution:

$$q(t) = c_1 \cos\left(\omega_0 t\right) + c_2 \sin\left(\omega_0 t\right).$$

Applying initial conditions $q(0) = Q_0$ and $i(0) = \dot{q}(0) = 0$:

$$c_1 = Q_0, c_2 = 0 \quad \rightarrow \quad q(t) = Q_0 \cos\left(\omega_0 t\right).$$

Differentiating to find the current and capacitor voltage:

$$i(t) = \frac{\mathrm{d}q}{\mathrm{d}t} = -Q_0\omega_0 \sin\left(\omega_0 t\right) \qquad \Delta V_C(t) = \frac{q(t)}{C} = \frac{Q_0}{C}\cos\left(\omega_0 t\right).$$

The energy in the circuit oscillates between the capacitor and the inductor, with total energy conserved:

$$U_C = \frac{q^2}{2C} = \frac{Q_0^2}{2C}\cos^2\left(\omega_0 t\right), \qquad U_L = \frac{1}{2}Li^2 = \frac{Q_0^2}{2C}\sin^2\left(\omega_0 t\right), \qquad U = U_C + U_L = \frac{Q_0^2}{2C}.$$

LC Series Circuit

For an LC series circuit with inductance L and capacitance C, the charge oscillates at the **natural angular frequency**:

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

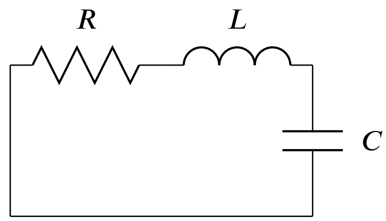
With initial conditions $q(0) = Q_0$ and $i(0) = 0$:

$$q(t) = Q_0 \cos\left(\omega_0 t\right) \qquad i(t) = -Q_0\omega_0 \sin\left(\omega_0 t\right) \qquad \Delta V_C(t) = \frac{Q_0}{C}\cos\left(\omega_0 t\right).$$

The total energy $U = Q_0^2/(2C)$ is conserved.

Inductor-Resistor-Capacitor (LRC) Series Circuits

An **LRC series circuit** has an inductor L, resistor R, and capacitor C all in series. The resistor dissipates energy, so unlike the LC circuit, oscillations are **damped** and eventually decay to zero.



With $i = \mathrm{d}q/\mathrm{d}t$, Kirchhoff's Loop Rule gives:

$$L\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{1}{C}q = 0.$$

Define the **damping coefficient** $\gamma = R/(2L)$ and **natural frequency** $\omega_0 = 1/\sqrt{LC}$:

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + 2\gamma\frac{\mathrm{d}q}{\mathrm{d}t} + \omega_0^2 q = 0.$$

The characteristic equation $r^2 + 2\gamma r + \omega_0^2 = 0$ has roots:

$$r = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}.$$

The three qualitatively distinct regimes depend on whether γ is less than, equal to, or greater than ω_0:

Underdamped ($\gamma < \omega_0$, i.e. $R^2 < 4L/C$)

The roots are complex: $r = -\gamma \pm i\omega_d$ where $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ is the **damped angular frequency**. The general solution is:

$$q(t) = e^{-\gamma t}\left(c_1\cos(\omega_d t) + c_2\sin(\omega_d t)\right).$$

In amplitude-phase form with $Q = \sqrt{c_1^2 + c_2^2}$ and $\varphi = \arctan(c_2/c_1)$:

$$q(t) = Qe^{-t/\tau}\cos(\omega_d t + \varphi), \qquad \tau = \frac{1}{\gamma} = \frac{2L}{R}.$$

The system oscillates at frequency $\omega_d < \omega_0$, with amplitude decaying exponentially. As $R \to 0$, $\omega_d \to \omega_0$ and we recover the undamped LC result.

Critically Damped ($\gamma = \omega_0$, i.e. $R^2 = 4L/C$)

The roots are real and equal: $r = -\gamma$. The general solution is:

$$q(t) = (c_1 + c_2 t)e^{-\gamma t}.$$

This is the boundary between oscillatory and monotonic behavior. The system returns to equilibrium as fast as possible *without oscillating*.

Overdamped ($\gamma > \omega_0$, i.e. $R^2 > 4L/C$)

The roots are real and distinct: $r_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$. The general solution is:

$$q(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

Both roots are negative, so $q(t) \to 0$ without oscillation, but more slowly than critical damping.

LRC Series Circuit

For an LRC series circuit with inductance L, resistance R, and capacitance C, define:

$$\gamma = \frac{R}{2L} \text{ (damping coefficient)} \qquad \omega_0 = \frac{1}{\sqrt{LC}} \text{ (natural frequency).}$$

The charge satisfies $\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + 2\gamma \frac{\mathrm{d}q}{\mathrm{d}t} + \omega_0^2 q = 0$. The solution depends on the **damping regime**:

- **Underdamped** ($\gamma < \omega_0$): $q(t) = Qe^{-\gamma t}\cos\left(\omega_d t + \varphi\right)$, where $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$.
- **Critically damped** ($\gamma = \omega_0$): $q(t) = \left(c_1 + c_2 t\right)e^{-\gamma t}$.
- **Overdamped** ($\gamma > \omega_0$): $q(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$, where $r_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$.

6 Alternating Current (AC) Circuits

6.1 AC Signals and Phasors

- An **alternating current** (AC) is a current that oscillates back and forth in a circuit, as opposed to a **direct current** (DC) which flows in only one direction.
- An **alternating voltage source** is a source of emf that produces an alternating current. A common example is the power outlet in your home, which provides an AC voltage.
- There are many types of **waveforms** for AC signals:
  - The **sinusoidal** waveform is the most common and is described by $V(t) = V_0 \sin(\omega t + \varphi)$, where V_0 is the amplitude, ω is the angular frequency, and φ is the phase.
  - A **rectangular wave** "jumps" between two levels with a *duty cycle* δ, for $0 < \delta < 1$.
    - The duty cycle is the fraction of one period in which the signal is "high".
    - A **square wave** has a duty cycle of $\delta = 0.5$.
  - A **triangular wave** linearly ramps up and down between two levels.
- There are three common ways to describe the strength of an AC signal:
  - **Amplitude** or **peak**: the maximum value. If $V(t) = V_0 \sin(\omega t + \varphi)$, then the amplitude is V_0.
  - **Peak-to-peak**: the difference between the maximum and minimum values. If $V(t) = V_0 \sin(\omega t + \varphi)$, then the peak-to-peak value is $V_{\text{pp}} = 2V_0$.
  - **Root mean square (RMS)**: typical way to quantify the "effective" value of an AC signal. Take the square of the signal, and find the mean value of the squared signal over one period, then take the square root of that mean:

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T V^2(t)\,\mathrm{d}t}.$$

    - For a sinusoidal signal, the RMS value is $V_{\text{rms}} = V_0/\sqrt{2}$.
    - For a rectangular wave with duty cycle δ, the RMS value is $V_{\text{rms}} = V_0\sqrt{\delta}$.
    - For a triangular wave, the RMS value is $V_{\text{rms}} = V_0/\sqrt{3}$.
- A sinusoidal signal $V_0 \cos(\omega t + \varphi)$ is completely described by:
  - its **amplitude** V_0,
  - its **angular frequency** ω (or equivalently, its frequency $f = \omega/(2\pi)$ or its period $T = 1/f$),
  - its **phase** or **phase shift** φ.

  The phase φ represents a horizontal shift of the signal. If $\varphi > 0$, then the signal is shifted to the left by $|\varphi|/\omega$ seconds, and if $\varphi < 0$, then the signal is shifted to the right by $|\varphi|/\omega$ seconds.

Phasors

To analyze a sinusoidal AC circuit, we can use **phasors**, which encode both the amplitude and phase of sinusoidal signals into complex numbers.

Visually, a phasor can be thought of as a vector in the complex plane.

> **Note**
>
> Since i is already used to represent current, we will use j to represent the imaginary number.

Take a sinusoidal signal $V(t) = V_0 \cos(\omega t)$. The phasor for this signal is $\tilde{V}(t) = V_0 e^{j\omega t}$. Realize that $V(t) = \mathrm{Re}\left(\tilde{V}(t)\right) = \mathrm{Re}\left(V_0 e^{j\omega t}\right)$, since by Euler's formula:

$$\mathrm{Re}\left(\tilde{V}(t)\right) = \mathrm{Re}\left(V_0 e^{j\omega t}\right) = V_0 \,\mathrm{Re}(\cos(\omega t) + j \sin(\omega t)) = V_0 \cos(\omega t).$$

Similarly, if our signal has a phase shift φ ($V(t) = V_0 \cos(\omega t + \varphi)$), the phasor is $\tilde{V}(t) = V_0 e^{j(\omega t + \varphi)}$. We can also write this phasor in **polar form** as $\tilde{V} = V_0 \angle\varphi$, writing both the amplitude V_0 and phase φ in a compact way. The term $\angle\varphi$ implicitly represents $e^{j(\omega t + \varphi)}$.

> **Phasor Representation of Sinusoidal Signals**
>
> A sinusoidal signal $V(t) = V_0 \cos(\omega t + \varphi)$ can be represented as its **phasor** $\tilde{V}(t)$ by adding an imaginary component $V_0 \sin(\omega t + \varphi) j$ to $V(t)$:
>
> $$\tilde{V}(t) = V_0 e^{j(\omega t + \varphi)}.$$
>
> The **polar form** of the phasor is $\tilde{V} = V_0 \angle\varphi$, where V_0 is the magnitude and φ is the *phase*, and the frequency ω is inferred from the context of the circuit. The term $\angle\varphi$ implicitly represents $e^{j(\omega t + \varphi)}$.

- Visually, the phasor $\tilde{V}$ is a "vector" of length V_0 that rotates counterclockwise in the complex plane at an angular velocity of ω. By projecting $\tilde{V}$ onto the real axis, we obtain the voltage $V(t)$ at any given time.
  - Two phasors $\tilde{u}(t) = u_0 \angle\varphi_1$ and $\tilde{y}(t) = v_0 \angle\varphi_2$ with the same angular velocity ω can be added together as follows:

    $$\begin{aligned} \tilde{u}(t) + \tilde{v}(t) &= u_0 e^{j\left(\omega t + \varphi_1\right)} + v_0 e^{j\left(\omega t + \varphi_2\right)} \\ &= \left(u_0 e^{j\varphi_1} + v_0 e^{j\varphi_2}\right) e^{j\omega t} \\ &= \left(u_0 \cos\varphi_1 + v_0 \cos\varphi_2 + j\left(u_0 \sin\varphi_1 + v_0 \sin\varphi_2\right)\right) e^{j\omega t} \\ &= \left(u_0 \cos\varphi_1 + v_0 \cos\varphi_2\right) e^{j\omega t} + j\left(u_0 \sin\varphi_1 + v_0 \sin\varphi_2\right) e^{j\omega t} \\ &= (\mathrm{Re}(\tilde{u}) + \mathrm{Re}(\tilde{v})) + j(\mathrm{Im}(\tilde{u}) + \mathrm{Im}(\tilde{v})). \end{aligned}$$

    On the complex plane, this is just the vector sum of the two phasors $\tilde{u}$ and $\tilde{v}$. That is, phasors can be added together using vector addition.

- Realize that if $\tilde{V}(t) = V_0 e^{j\omega t}$, then $\mathrm{d}\tilde{V}/\,\mathrm{d}t = j\omega V_0 e^{j\omega t}$. This means that taking the time-derivative of a phasor corresponds to multiplying it by $j\omega$. Since $j = e^{j\pi/2}$, the phase is shifted counterclockwise by $\pi/2$ radians (i.e. 90 degrees) when we take the time derivative of a phasor. In general,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(V_0\,\angle\varphi\big) = j\omega V_0\,\angle\varphi = V_0\omega\,\angle\Big(\varphi + \frac{\pi}{2}\Big).$$

  - Conversely, *integrating* a phasor corresponds to *dividing* by $j\omega$, i.e. phase shift *clockwise* by $\pi/2$:

$$\int V_0\,\angle\varphi\,\mathrm{d}t = \frac{V_0}{j\omega}\,\angle\varphi = \frac{V_0}{\omega}\,\angle\Big(\varphi - \frac{\pi}{2}\Big).$$

    This comes from the fact that $1/j = -j = -e^{j\pi/2}$.

- If two phasors have the same phase shift, then they are said to be **in phase**.
  - If a phasor $\tilde{u}$ is shifted counterclockwise by θ relative to another phasor $\tilde{v}$, then $\tilde{u}$ is said to **lead** $\tilde{v}$ by θ, and $\tilde{v}$ is said to **lag** $\tilde{u}$ by θ, and if $\theta \not\equiv 0 \pmod{2\pi}$, then $\tilde{u}$ and $\tilde{v}$ are said to be **out of phase**.
  - If $\tilde{u}$ and $\tilde{v}$ have a phase difference of π radians, then $\tilde{u}$ and $\tilde{v}$ are *perfectly* out of phase.

6.2 Reactance and Impedance

Reactance

The **reactance** X of a circuit element measures how much it opposes the flow of *alternating* current.

If $V(t)$ and $I(t)$ have an absolute phase difference of exactly θ, then the reactance X of the circuit element is defined as:

$$X = \frac{V_0}{I_0} \sin\theta.$$

If $V(t)$ and $I(t)$ are exactly 90° out of phase, then $\sin\theta = 1$, and the reactance is simply the ratio of the voltage amplitude to the current amplitude: $X = V_0/I_0$, so we get an Ohm-like relationship:

$$X = \frac{V_0}{I_0} \quad \rightarrow \quad V_0 = I_0 X$$

- **Reactance of a resistor.** Let an AC circuit provide a voltage source given by $V_R(t) = V_0\cos(\omega t)$. If the circuit contains a resistor with resistance R, then by Ohm's Law, the current through the resistor is:

  $$I_R(t) = \frac{V_R(t)}{R} = \frac{V_0}{R}\cos(\omega t).$$

  Since $V_R(t)$ and $I(t)$ are in phase, the reactance of a resistor is 0 ($\sin 0 = 0$). This means that a resistor does not oppose the flow of *alternating* current, i.e. it doesn't prevent the current from oscillating back and forth in the circuit.

  This can be thought of as the fact that a resistor dissipates energy as heat and never returns any power back to the circuit, unlike capacitors or inductors.

  Since $V_R(t)$ is in phase with $i(t)$, their phasors $\tilde{V}_R$ and $\tilde{\imath}$ are also in phase.

- **Reactance of a capacitor.** Let an AC circuit provide a potential difference across a capacitor C given by $V_C(t) = V_0\cos(\omega t)$. Then, the charge stored on a capacitor at time t is given by $q(t) = CV_C(t)$, so:

  $$i(t) = \frac{\mathrm{d}q}{\mathrm{d}t} = C\frac{\mathrm{d}V_C}{\mathrm{d}t} = -CV_0\omega\sin(\omega t) = CV_0\omega\cos\left(\omega t + \frac{\pi}{2}\right) \rightarrow I_0 = CV_0\omega.$$

  Since $V_C(t)$ and $i(t)$ are 90° out of phase, the reactance of a capacitor is $X_C = V_0/I_0 = 1/(C\omega)$:

  Capacitive Reactance

  The **capacitive reactance** X_C of a capacitor with capacitance C in an AC circuit with angular frequency ω is:

  $$X_C = \frac{1}{C\omega}.$$

  Realize that $i(t)$ is exactly 90° *ahead* of $V_C(t)$. Since $\tilde{\imath}$ is in phase with $\tilde{V}_R$, in a circuit with both a resistor and a capacitor, $\tilde{V}_R$ *leads* $\tilde{V}_C$ by 90° (voltage across resistor leads voltage across capacitor by 90°).

- **Reactance of an inductor.** The voltage across an inductor is given by $V_L(t) = L\,\mathrm{d}i/\,\mathrm{d}t$. Let a voltage source provide an AC voltage $V_L(t) = V_0\cos(\omega t)$ across an inductor with inductance L, so:

$$\begin{aligned} L\frac{\mathrm{d}i}{\mathrm{d}t} = V_0\cos(\omega t) \quad \rightarrow \quad \frac{\mathrm{d}i}{\mathrm{d}t} &= \frac{V_0}{L}\cos(\omega t) \\ i(t) &= \frac{V_0}{L}\int \cos(\omega t)\,\mathrm{d}t \\ &= \frac{V_0}{L\omega}\sin(\omega t) = \frac{V_0}{L\omega}\cos\left(\omega t - \frac{\pi}{2}\right) \rightarrow I_0 = \frac{V_0}{L\omega}. \end{aligned}$$

Since $V_L(t)$ and $i(t)$ are 90° out of phase, the reactance of an inductor is $X_L = V_0/I_0 = L\omega$:

Inductive Reactance

The **inductive reactance** X_L of an inductor with inductance L in an AC circuit with angular frequency ω is:

$$X_L = L\omega.$$

Realize that $V_L(t)$ is exactly 90° *ahead* of $i(t)$, so $\tilde{V}_L$ *leads* $\tilde{\imath}$ by 90°. Since $\tilde{\imath}$ is in phase with $\tilde{V}_R$, in a circuit with both a resistor and an inductor, $\tilde{V}_L$ *leads* $\tilde{V}_R$ by 90° (voltage across inductor leads voltage across resistor by 90°).

In a circuit with a resistor, capacitor, and inductor, the voltage across the inductor leads the voltage across the resistor by 90°, which leads the voltage across the capacitor by 90°.
- In other words, $\tilde{V}_L$ leads $\tilde{V}_R$ by 90°, which leads $\tilde{V}_C$ by 90°.
- This also means that $\tilde{V}_L$ leads $\tilde{V}_C$ by 180°, i.e. $\tilde{V}_L$ and $\tilde{V}_C$ are perfectly out of phase.

Impedance

The **impedance** Z of a circuit element is a generalization of resistance that applies to AC circuits. It is defined as the ratio of the voltage phasor to the current phasor:

$$Z = \frac{\tilde{V}}{\tilde{I}} \quad \rightarrow \quad \tilde{V} = Z\tilde{I}.$$

Since $\tilde{V}$ and $\tilde{I}$ are complex functions, the impedance Z is a complex number! In fact, the impedance of a circuit element can be expressed in terms of its resistance R and reactance X as follows:

$$Z = R + jX.$$

Thus, we can redefine the reactance X of a circuit element as the imaginary part of its impedance:

$$X = \mathrm{Im}(Z).$$

- **Impedance of a resistor.** Let an AC circuit provide a voltage source given by $V(t) = V_0\cos(\omega t + \varphi)$ across a resistor with resistance R, so the phasor for this voltage is $\tilde{V}_R = V_0\angle\varphi$. Then:

$$\tilde{\imath} = \frac{\tilde{V}_R}{R} = \frac{V_0}{R}\angle\varphi.$$

By the definition of impedance:

$$Z_R = \frac{\tilde{V}_R}{\tilde{\imath}} = \frac{V_0 \angle\varphi}{V_0/R \angle\varphi} = R.$$

Since Z_R has no imaginary part, we prove that reactance of a resistor is 0.

- **Impedance of a capacitor.** Let an AC circuit provide a potential difference across a capacitor C given by $V_C(t) = V_0 \cos(\omega t + \varphi)$, so the phasor for this voltage is $\tilde{V}_C = V_0 \angle\varphi$. Then, the charge stored on a capacitor at time t is given by $q(t) = CV_C(t)$, so:

$$\tilde{\imath} = \frac{\mathrm{d}q}{\mathrm{d}t} = C\frac{\mathrm{d}\tilde{V}_C}{\mathrm{d}t} = C\frac{\mathrm{d}}{\mathrm{d}t}V_0 e^{j(\omega t+\varphi)} = CV_0 j\omega e^{j(\omega t+\varphi)} = CV_0 j\omega \angle\varphi.$$

  Then the impedance of the capacitor is:

$$Z_C = \frac{\tilde{V}_C}{\tilde{\imath}} = \frac{V_0 \angle\varphi}{CV_0 j\omega \angle\varphi} = \frac{1}{j\omega C} = -\frac{j}{\omega C}.$$

  Since Z_C has no real part, we prove that the resistance of a capacitor is 0 (it dissipates no energy) and the reactance of a capacitor is $-1/(\omega C)$.[9]

- **Impedance of an inductor.** Let a voltage source provide an AC voltage $V_L(t) = V_0 \cos(\omega t + \varphi)$ across an inductor with inductance L, so the phasor for this voltage is $\tilde{V}_L = V_0 \angle\varphi$. Then:

$$V_0 \angle\varphi = L\frac{\mathrm{d}\tilde{\imath}}{\mathrm{d}t} \quad \to \quad \frac{\mathrm{d}\tilde{\imath}}{\mathrm{d}t} = \frac{V_0}{L}\angle\varphi$$
$$\tilde{\imath} = \frac{V_0}{L}\int \angle\varphi \,\mathrm{d}t = \frac{V_0}{L}\int e^{j(\omega t+\varphi)}\,\mathrm{d}t = \frac{V_0}{Lj\omega}e^{j(\omega t+\varphi)} = \frac{V_0}{Lj\omega}\angle\varphi.$$

  Then the impedance of the inductor is:

$$Z_L = \frac{\tilde{V}_L}{\tilde{\imath}} = \frac{V_0 \angle\varphi}{V_0/(Lj\omega)\angle\varphi} = j\omega L.$$

  Since Z_L has no real part, we prove that the resistance of an inductor is 0 (it dissipates no energy) and the reactance of an inductor is ωL.

If a circuit has an AC power source $\tilde{V}$ and has an impedance Z, then the current phasor $\tilde{\imath}$ in the circuit is given by:

$$\tilde{\imath} = \frac{\tilde{V}}{Z}.$$

[9] Use the absolute value to obtain $X_C = 1/(\omega C)$.

6.3 LRC Circuits with AC Sources

Impedance in a series LRC Circuit

Let an AC circuit provide a voltage source $V(t) = V_0 \cos(\omega t + \varphi)$ across a series LRC circuit with resistance R, inductance L, and capacitance C. Then, the impedance Z of the circuit is:

$$Z = R + j\left(X_L - X_C\right) = R + j\left(\omega L - \frac{1}{\omega C}\right).$$

- The magnitude of the impedance, often called the **impedance magnitude**, is:

$$|Z| = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}.$$

  The current amplitude I_0 in the circuit is given by the ratio of the voltage amplitude to the impedance magnitude:

$$I_0 = \frac{V_0}{|Z|} = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}.$$

- The phase angle θ of the impedance is given by its *complex argument* $\theta = \arg Z$, where:

$$\tan\theta = \frac{X_L - X_C}{R}.$$

  In which case, the current phasor $\tilde{\imath}$ can be expressed in terms of the voltage phasor $\tilde{V}$ and the impedance Z as follows:

$$\tilde{\imath} = \frac{\tilde{V}}{Z} = \frac{V_0 \angle\varphi}{R + j\left(X_L - X_C\right)} = \frac{V_0}{|Z|} \angle(\varphi - \theta) = I_0 \angle(\varphi - \theta).$$

  That is, **the current lags the voltage by an angle of θ**.

Potential Difference across a series LRC Circuit

Let an AC circuit provide emf $\mathcal{E}(t) = V_0 \cos(\omega t + \varphi)$ across a series LRC circuit with resistance R, inductance L, and capacitance C. Then, the potential difference across the circuit is:

$$V(t) = \sqrt{V_R^2 + \left(V_L - V_C\right)^2} \cos(\omega t + \varphi - \theta),$$

where:

- $V_R = I_0 R$, $V_L = I_0 X_L$, and $V_C = I_0 X_C$ are the voltage amplitudes across the resistor, inductor, and capacitor, respectively.
- $\theta = \arctan\left(\left(V_L - V_C\right)/V_R\right)$ is the phase angle of the impedance.

6.4 Power in AC Circuits

Consider a circuit component in an AC circuit with potential difference $\tilde{V}$ and current $\tilde{\imath}$, where $\tilde{V}$ and $\tilde{\imath}$ are out of phase by φ. That is, if $V(t) = V_0 \cos(\omega t)$, then $I(t) = I_0 \cos(\omega t + \varphi)$. The instantaneous power $P(t)$ absorbed by the circuit component is given by:

$$P(t) = V(t)I(t) = \big(V_0 \cos(\omega t)\big)\big(I_0 \cos(\omega t + \varphi)\big) = V_0 I_0 \cos(\omega t)\cos(\omega t + \varphi).$$

Then, the **average power** P_{avg} delivered to the circuit over one period is given by:

$$P_{\text{avg}} = \frac{1}{T}\int_0^T P(t)\,\mathrm{d}t = \frac{1}{T}\int_0^T V_0 I_0 \cos(\omega t + \varphi)\cos(\omega t)\,\mathrm{d}t = \frac{1}{2}V_0 I_0 \cos\varphi.$$

Realize that if the voltage and current are in phase (as in a resistor), then $\varphi = 0$, and the average power is maximized at $P_{\text{avg}} = \frac{1}{2}V_0 I_0$. That is, resistors absorb the maximum possible power from an AC source.

If the voltage and current are out of phase by exactly 90° (like in an inductor or capacitor), then $\varphi = \pi/2$, and the average power is $P_{\text{avg}} = 0$ (no net energy is delivered to the circuit over one period)

Power Factor

The **power factor** of a circuit component is $\cos\varphi$, which is the cosine of the phase difference φ between the voltage and current phasors:

$$\text{power factor} = \cos\varphi.$$

The power factor represents the fraction of the maximum possible power that is actually delivered to the circuit component. For example, if $\varphi = 60°$, then $\cos\varphi = 0.5$, so only half of the maximum possible power is delivered to the circuit component.

Using V_{rms} and I_{rms}, the 1/2 term from average power vanishes:

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\varphi.$$

For a resistor, the power factor is $\cos\varphi = 1$, so:

$$\underset{\text{resistor}}{P_{\text{avg}}} = V_{\text{rms}} I_{\text{rms}} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}.$$

Power Absorbed by a Circuit Element in an AC Circuit

The **average power** P_{avg} absorbed by a circuit element in an AC circuit with maximum potential difference V_0, maximum current I_0, and phase difference φ between the potential difference across the element and current is given by:

$$P_{\text{avg}} = \frac{1}{2}V_0 I_0 \cos\varphi = V_{\text{rms}} I_{\text{rms}} \cos\varphi,$$

where V_{rms} and I_{rms} are the root mean square values of the potential difference and current, respectively.

Complex Power

For an AC voltage source with voltage phasor $\tilde{V}$ and current phasor $\tilde{\imath}$, the **complex power** S delivered to the circuit is defined as:

$$S = \tilde{V}\tilde{\imath}^*,$$

where $\tilde{\imath}^*$ is the complex conjugate of $\tilde{\imath}$. The complex power can be expressed in terms of the **active power** P and the **reactive power** Q as follows:

$$S = P + jQ.$$

- The reactive power Q represents the power that is alternately absorbed and released by the circuit component, and it does not contribute to the net energy delivered to the circuit over one period.

  Reactive or "imaginary" power Q is the rate at which energy is stored and released by the circuit component, while active or "real" power P is the rate at which energy is dissipated as heat or delivered to a load.

The other equations for power have analogous forms:

$$\begin{aligned} S &= \tilde{V}\tilde{\imath}^* &\sim\quad P &= VI \\ S &= |\tilde{\imath}|^2 Z &\sim\quad P &= I^2 R \\ S &= \frac{|\tilde{V}|^2}{Z^*} &\sim\quad P &= \frac{V^2}{R} \end{aligned}$$

A nice result of this is that the components of complex power can be expressed as follows:

$$P = \mathrm{Re}(S) = |\tilde{\imath}|^2\,\mathrm{Re}(Z) = |\tilde{\imath}|^2 R = \frac{|\tilde{V}|^2}{|Z|^2}R, \qquad Q = \mathrm{Im}(S) = |\tilde{\imath}|^2\,\mathrm{Im}(Z) = |\tilde{\imath}|^2 X = \frac{|\tilde{V}|^2}{|Z|^2}X.$$

The magnitude of complex power $|S| = \sqrt{P^2 + Q^2}$ is called **apparent power**. An AC circuit with any configuration of resistors, inductors, and capacitors have a constant apparent power! That is, S "rotates" at an angular frequency ω in the complex plane, but its magnitude $|S|$ remains constant. The average power P_{avg} of an AC circuit with phase difference[10] φ is given by:

$$\begin{aligned} P_{\mathrm{avg}} &= \mathrm{Re}(S) \\ V_{\mathrm{rms}} I_{\mathrm{rms}} \cos\varphi &= |S|\cos\varphi. \end{aligned}$$

That is, $|S| = V_{\mathrm{rms}} I_{\mathrm{rms}}$.

Apparent Power of AC Generator

The **apparent power** $|S|$ generated by an AC voltage source with RMS voltage V_{rms} and RMS current I_{rms} is:

$$|S| = V_{\mathrm{rms}} I_{\mathrm{rms}}.$$

[10] Phase of voltage relative to current

6.5 Resonance in AC Circuits

An AC circuit acheives **resonance** when the current in the circuit is maximized and the impedance is maximized. Since impedance is a function of the angular frequency ω of the AC source, the **resonant angular frequency** ω_r of an AC circuit is the frequency at which resonance occurs.

A series LRC circuit is at resonance when the inductive reactance and capacitive reactance are equal, such that the impedance of the circuit is purely resistive (i.e. $X_L - X_C = 0 \rightarrow Z = R$). In such case, *all* power from the power source is delivered to the resistor, and *none* of the power is stored in the inductor or capacitor.

To find the resonant angular frequency, we set $X_L = X_C$:

$$X_L = X_C \quad \rightarrow \quad \omega L = \frac{1}{\omega C} \quad \rightarrow \quad \omega^2 = \frac{1}{LC} \quad \rightarrow \quad \omega_r = \frac{1}{\sqrt{LC}}.$$

Resonant Frequency of a series LRC Circuit

The **resonant angular frequency** ω_r of a series LRC circuit with inductance L and capacitance C is:

$$\omega_r = \frac{1}{\sqrt{LC}}.$$

Thus, the **resonant frequency** f_r is:

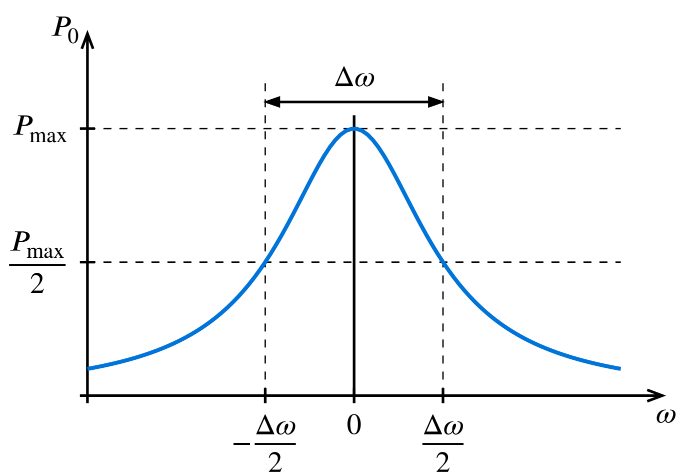
$$f_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi\sqrt{LC}}.$$

At resonance, the effect of the inductor and capacitor cancel each other out!

Quality Factor

The **quality factor** Q of an AC circuit is a measure of how precise the AC frequency must be to acheive resonance. It is defined as the ratio of the resonant frequency ω_r to the *bandwidth* $\Delta\omega$ of the circuit (see figure to the right):

$$Q = \frac{\omega_r}{\Delta\omega}.$$



The bandwidth $\Delta\omega$ is the range of frequencies around the resonant frequency for which the power supplied to the circuit is at least half of the maximum power at resonance. This is called *full width at half maximum*.

For a series LRC circuit, find the ω at which

AC Transformers

7 Electromagnetic Waves

7.1 Displacement Current

Recall Ampere's Law

Displacement Current

Let S be a surface with a time-varying electric flux Φ_E. The **displacement current** I_{disp} passing through S is defined as:

$$I_{\text{disp}} = \varepsilon_0 \frac{\partial \Phi_E}{\partial t} = \varepsilon_0 \frac{\partial}{\partial t} \iint_S \vec{E} \cdot \mathrm{d}\vec{A}.$$

Ampere's Law with Maxwell's Correction

Let S be a surface bounded by a closed curve C. Let I_{enc} be the current passing through S and Φ_E be the electric flux through S. Then, the **Ampere-Maxwell** law states that:

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{\ell} = \mu_0 \left(I_{\text{enc}} + I_{\text{disp}} \right) = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{\partial \Phi_E}{\partial t} \right).$$

This is the fourth of Maxwell's equations.

Differential Form of Ampere-Maxwell Law

Using the definition of Φ_E gives:

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{\ell} = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{\partial}{\partial t} \iint_S \vec{E} \cdot \mathrm{d}\vec{A} \right).$$

We can also rewrite I_{enc} as a surface integral of the current density $\vec{J}$ over S:

$$\begin{aligned} \oint_C \vec{B} \cdot \mathrm{d}\vec{\ell} &= \mu_0 \left(\iint_S \vec{J} \cdot \mathrm{d}\vec{A} + \varepsilon_0 \frac{\partial}{\partial t} \iint_S \vec{E} \cdot \mathrm{d}\vec{A} \right) \\ &= \iint_S \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \cdot \mathrm{d}\vec{A}. \end{aligned}$$

By Stokes' theorem, the left-hand side can be rewritten as a surface integral of the curl of $\vec{B}$:

$$\iint_S \left(\vec{\nabla} \times \vec{B} \right) \cdot \mathrm{d}\vec{A} = \iint_S \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \cdot \mathrm{d}\vec{A}.$$

The integrands must be equal, so we have the differential form of Ampere-Maxwell law:

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right).$$

Since $\vec{D} = \varepsilon_0 \vec{E}$ is the *displacement field*, we can also realize that:

$$\varepsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{\partial \vec{D}}{\partial t} = \vec{J}_{\mathrm{disp}},$$

so the Ampere-Maxwell law can also be written as:

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \vec{J}_{\mathrm{disp}} \right),$$

where $\vec{J}_{\mathrm{disp}}$ is the **displacement current density**.

Ampere-Maxwell Law (Differential Form)

Let S be a surface over which the electric field $\vec{E}$ is changing with time, and let $\vec{J}$ be the current density at each point on S. Then, the **Ampere-Maxwell law in differential form** states that for any point on S:

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \vec{J}_{\mathrm{disp}} \right) = \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right).$$

7.2 Deriving the Wave Equation

In a vaccum, there are no free charges or currents, so Maxwell's equations reduce to:

$$\vec{\nabla} \cdot \vec{E} = 0$$
$$\vec{\nabla} \cdot \vec{B} = 0$$
$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
$$\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

Taking the curl of both sides of the third equation, we have:

$$\begin{aligned}
\vec{\nabla} \times \left(\vec{\nabla} \times \vec{E}\right) &= \vec{\nabla} \times \left(-\frac{\partial \vec{B}}{\partial t}\right) \\
&= -\frac{\partial}{\partial t}\left(\vec{\nabla} \times \vec{B}\right) \\
&= -\frac{\partial}{\partial t}\left(\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right) \\
&= -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}.
\end{aligned}$$

Using the vector identity $\vec{\nabla} \times \left(\vec{\nabla} \times \vec{E}\right) = \vec{\nabla}\left(\vec{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E}$, we can rewrite the left-hand side as

$$\vec{\nabla} \times \left(\vec{\nabla} \times \vec{E}\right) = \vec{\nabla}\left(\vec{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E} = -\nabla^2 \vec{E}.$$

Thus, we have the wave equation for the electric field in a vacuum:

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}.$$

7.3 Energy and Momentum of Electromagnetic Waves

7.4 Electromagnetic Spectrum